

机器人学

第三章：位姿描述和齐次变换

厦门大学信息学院

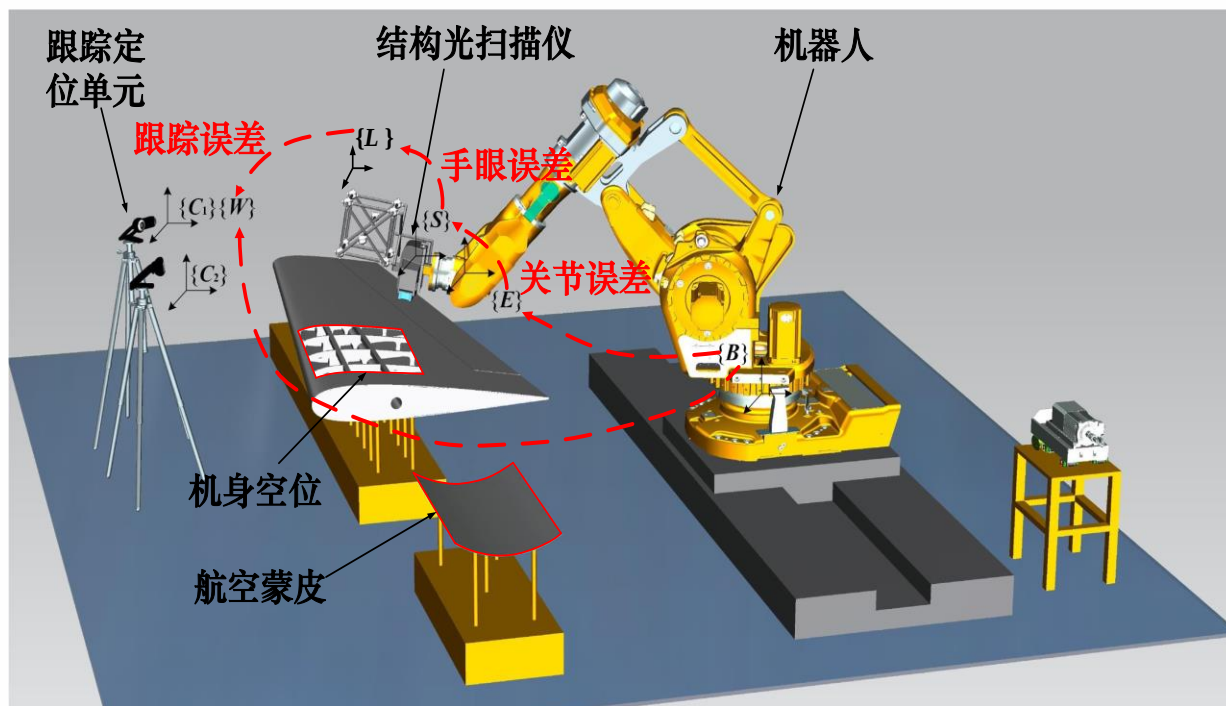
人工智能系

晁飞副教授



位姿描述—机器人铣削修边应用场景

飞机蒙皮加工系统：机器人、扫描仪、跟踪仪、工具、工件



位姿描述—机器人操作（磨抛）应用场景

大叶片加工工艺：模锻——铣削——变形矫正——磨抛



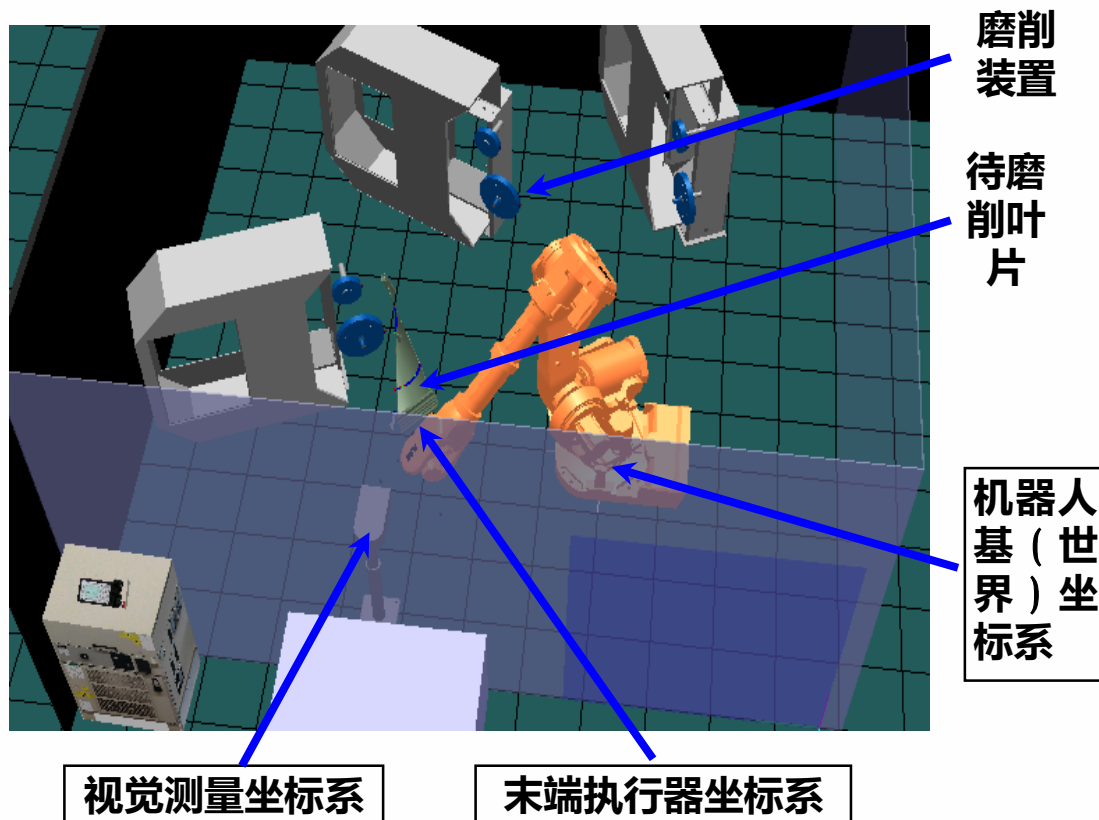
手工磨抛



机器人磨抛

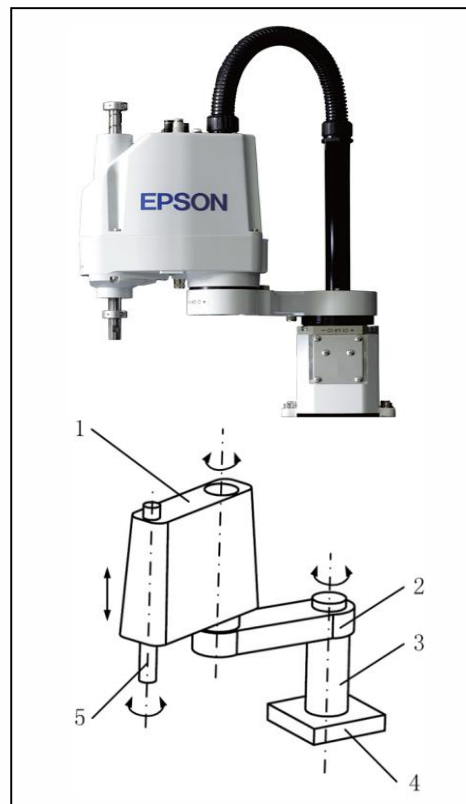
位姿描述—机器人操作（磨抛）应用场景

机器人磨抛涉及的坐标系：



本章提纲

- 3.1 刚体位姿描述
- 3.2 齐次坐标和齐次变换
- 3.3 运动算子
- 3.4 变换矩阵的运算
- 3.5 欧拉角与RPY角
- 3.6 旋转变换通式
- 3.7 位姿的综合
- 3.8 计算的复杂性



3.1 刚体位姿描述

- 研究机器人操作臂的运动：涉及各连杆位姿关系、连杆与周围环境（操作对象和障碍物）的关系。我们把操作臂的**各个连杆、操作对象、工具、工件和障碍物**都当成刚体。
- 刚体位姿描述：**齐次变换（矩阵）、矢量法、四元数**

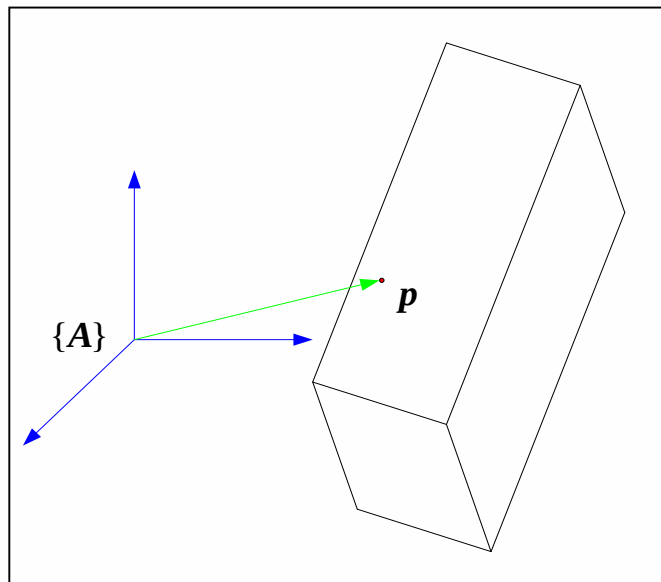
- ◆ 位姿 (Location) = 位置 (Position) + 姿态 (Orientation)
- ◆ 齐次变换法
 - 将运动、变换和映射与矩阵运算联系起来；
 - 在操作臂运动/动力学、机器人控制算法、计算机图学、视觉信息处理、手-眼建模标定都有广泛应用。
- ◆ 描述刚体在坐标系中的相对位姿（描述）； Description
- ◆ 表示刚体运动前后位姿描述的变换（算子）； Operator
- ◆ 把点从一个坐标系映射到另一个坐标系（映射）； Mapping

3.1 位置的描述（位置矢量）

□ 在坐标系 $\{A\}$ 中，空间任意一点可表示为列矢量 ${}^A\mathbf{p}$

$${}^A\mathbf{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}$$

直角坐标系



□ 符号表示：矢量/向量等（小写粗体）、矩阵/坐标系/群等（大写粗体）、变量/角度等（小写非斜体）

3.1 旋转矩阵与旋转群

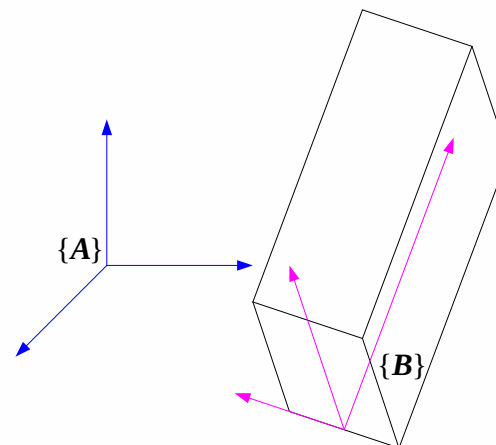
□ 坐标系 {B} 相对于坐标系 {A} 的旋转矩阵

$${}^A_B\mathbf{R} = \begin{bmatrix} {}^A\mathbf{x}_B & {}^A\mathbf{y}_B & {}^A\mathbf{z}_B \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

${}^A_B\mathbf{R}$ 是正交矩阵，满足如下关系：

$${}^A\mathbf{x}_B \cdot {}^A\mathbf{x}_B = {}^A\mathbf{y}_B \cdot {}^A\mathbf{y}_B = {}^A\mathbf{z}_B \cdot {}^A\mathbf{z}_B = 1$$

$${}^A\mathbf{x}_B \cdot {}^A\mathbf{y}_B = {}^A\mathbf{y}_B \cdot {}^A\mathbf{z}_B = {}^A\mathbf{z}_B \cdot {}^A\mathbf{x}_B = 0$$



□ 所以旋转矩阵中9个元素只有3个独立变量

□ 还满足： ${}^A_B\mathbf{R}^{-1} = {}^A_B\mathbf{R}^T$ ； $\det({}^A_B\mathbf{R}) = +1$

上标T表示转置，det表示行列式符号

刚体-1

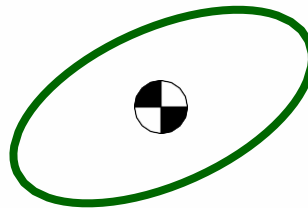
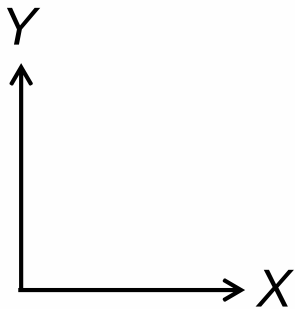
□ 一个刚体(Rigid body)的状态该如何描述？

◆ 平面：

移动？ DOFs、转动？ DOF

 Degree of freedom

$\{W\}$ world frame



刚体-1

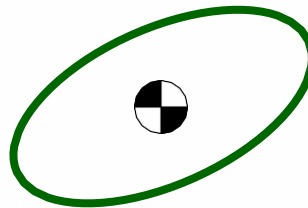
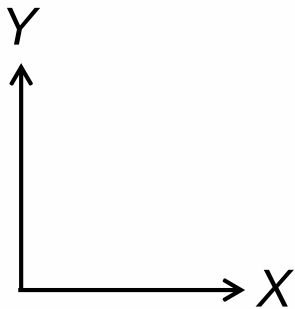
□ 一个刚体(Rigid body)的状态该如何描述？

◆ 平面：

移动 2 DOFs、转动 1 DOF

 Degree of freedom

$\{W\}$ world frame



刚体 -2

□ 一个刚体(Rigid body)的状态该如何描述？

◆ 平面：

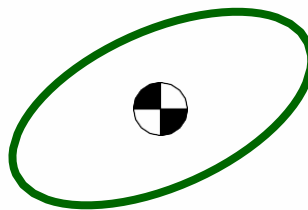
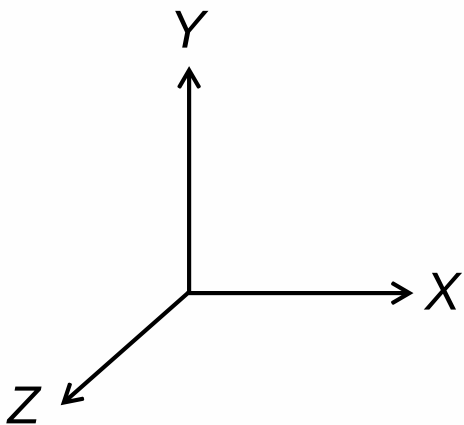
移动 2 DOFs、转动 1 DOF

—— Degree of freedom

◆ 空间：

移动 ? DOFs、转动 ? DOFs

$\{W\}$ world frame



刚体 -2

□ 一个刚体(Rigid body)的状态该如何描述？

◆ 平面：

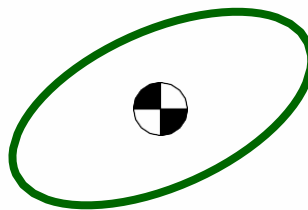
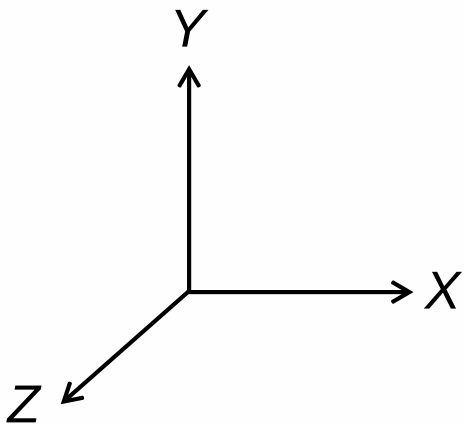
移动 2 DOFs、转动 1 DOF

Degree of freedom

◆ 空间：

移动 3 DOFs、转动 3 DOFs

$\{W\}$ world frame

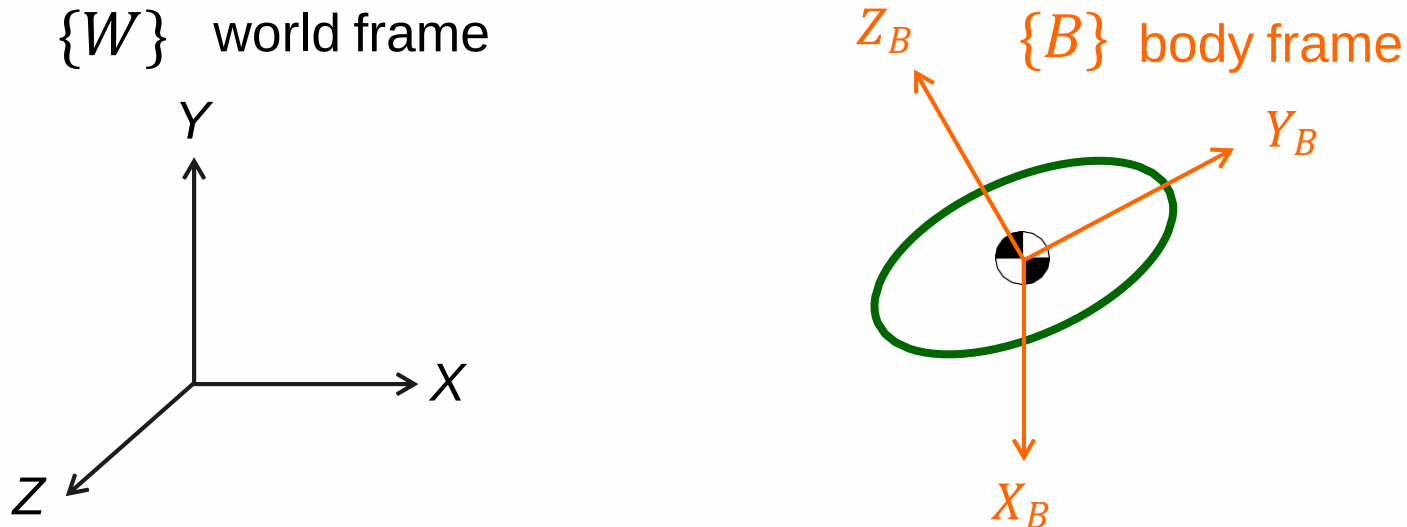


刚体-3

□ 该如何整合表达刚体的状态？

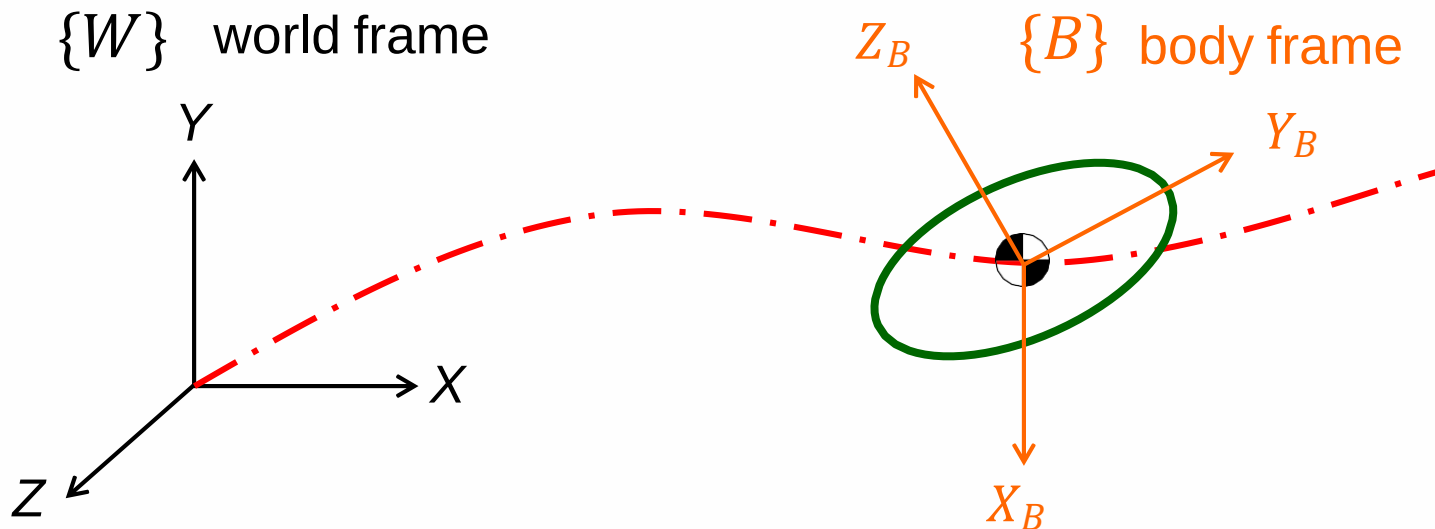
⇒ 在刚体(Rigid body)上建立frame，常建立在质心上

- ◆ 移动：由body frame 的原点位置判定
- ◆ 转动：由body frame的姿态判定



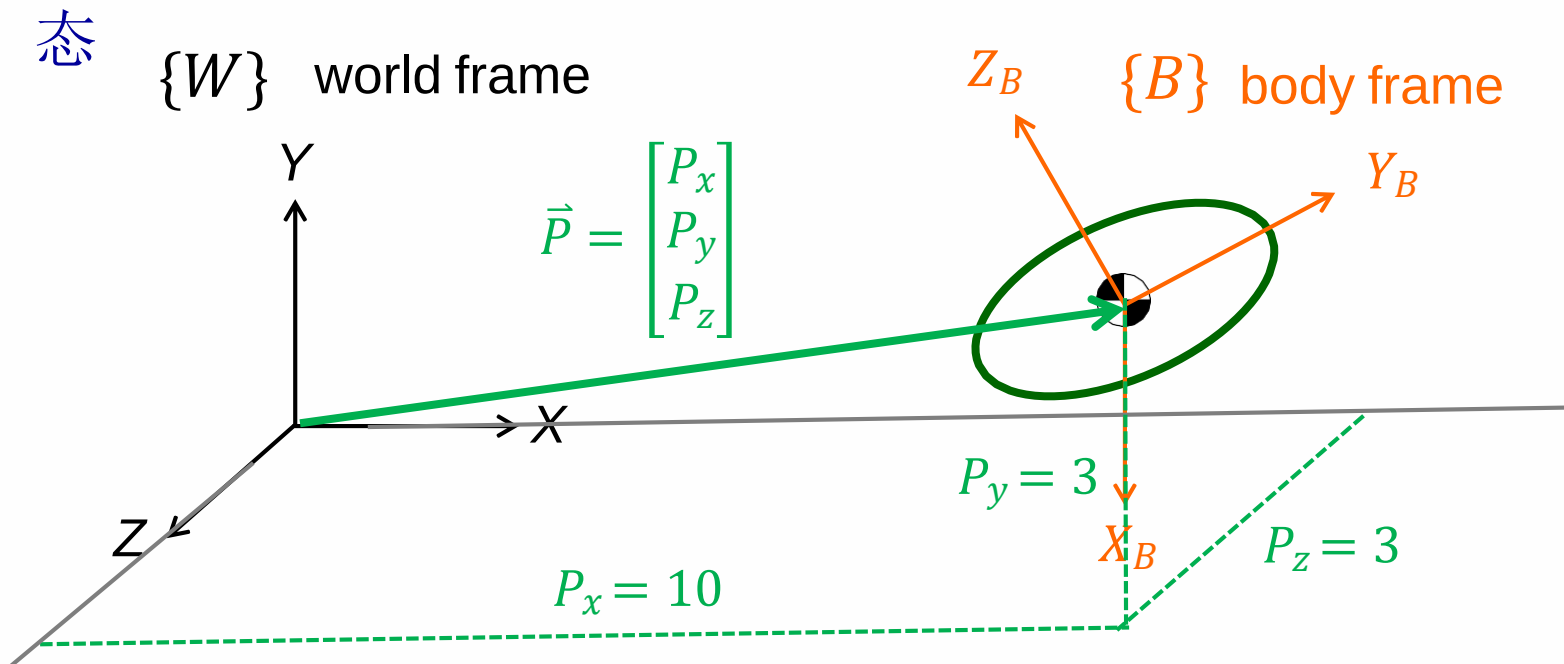
导读 -4

- 一个刚体(Rigid body)的“运动”状态该如何描述？
 - ◆ 利用各个DOF的微分，将位移和姿态（displacement / orientation）转换到速度（velocity）和加速度（acceleration）等运动状态



移动 -1

- 移动：以向量（vector） $\vec{P}$ 来描述 $\{B\}$ 的原点相对于 $\{A\}$ 的状态



- Ex: $\vec{P} = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} 10 \\ 3 \\ 3 \end{bmatrix}$

移动 -2

□ 向量可表达空间关系的两个方式

- ◆ A position in space (i.e., position vector)

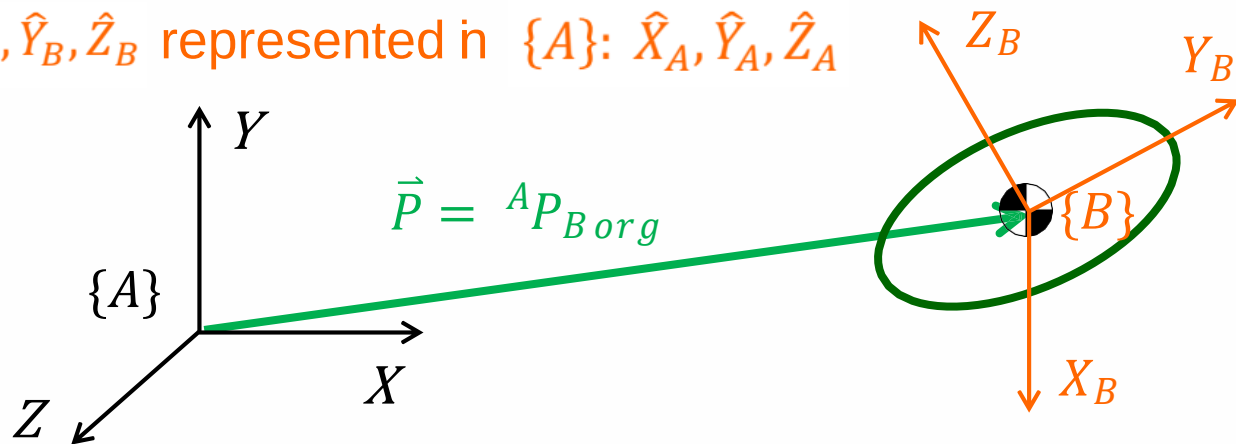
如同前一页内容，以此方式描述body frame原点

$$\vec{P} = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = {}^A P_{B\text{org}} = \text{origin of } \{B\} \text{ represented in } \{A\}$$

- ◆ A vector (i.e., displacement, frame basis)

以此方式表达body frame上主轴的方向

$\{B\}$: $\hat{X}_B, \hat{Y}_B, \hat{Z}_B$ represented in $\{A\}$: $\hat{X}_A, \hat{Y}_A, \hat{Z}_A$

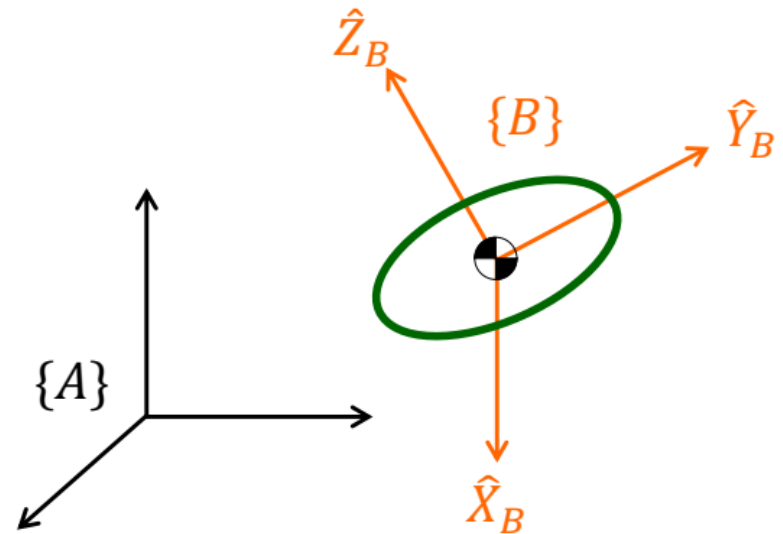


转动 -1

- 转动：描述{B}相对于{A}之姿态---旋转矩阵

$$\begin{aligned} \overset{\substack{\uparrow \\ \text{B relative} \\ \text{to A}}}{{}^A_B R} &= \begin{bmatrix} | & | & | \\ {}^A \hat{X}_B & {}^A \hat{Y}_B & {}^A \hat{Z}_B \\ | & | & | \end{bmatrix} \quad \text{"column vector"} \\ &= \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} \\ &\quad \text{"direct cosines"} \end{aligned}$$

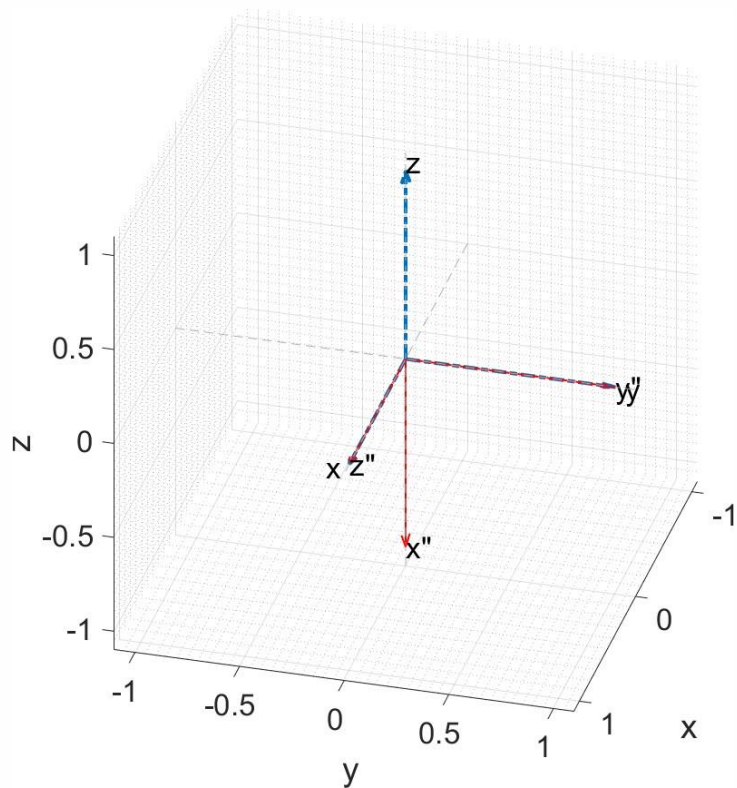
R的三个columns即为frame {B}的basis: $\hat{X}_B, \hat{Y}_B, \hat{Z}_B$ (由{A}看)



转动 -2

$$= \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix}$$

□ Ex: $\{B\}$ 相对于 $\{A\}$ 之姿态 ${}^A_R = ?$



蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$

$\{B\}$ 的 x'' 轴为 $\{A\}$ 的 z 轴反向 $\Rightarrow {}^A\hat{X}_B = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$

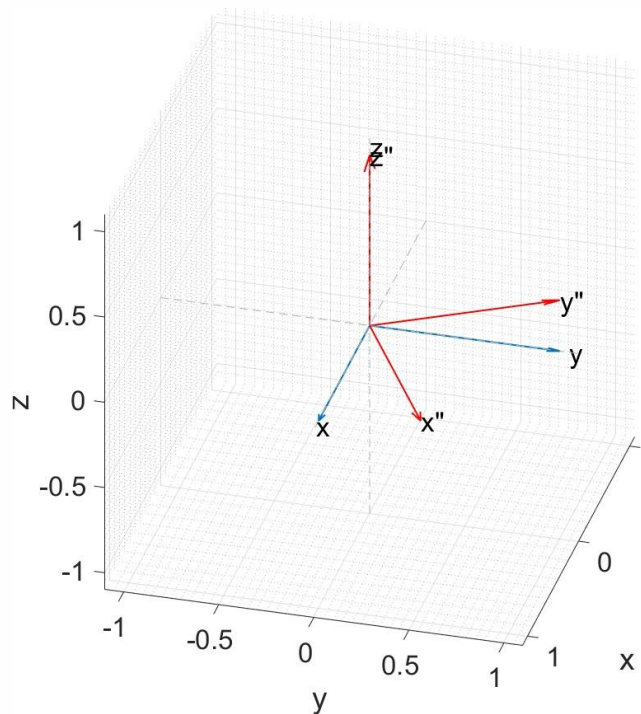
$\{B\}$ 的 y'' 轴与 $\{A\}$ 的 y 轴重迭 $\Rightarrow {}^A\hat{Y}_B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

$\{B\}$ 的 z'' 轴与 $\{A\}$ 的 x 轴重迭 $\Rightarrow {}^A\hat{Z}_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

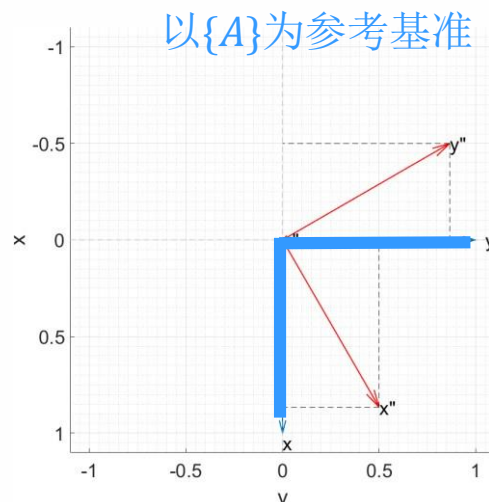
因此, $\{B\}$ 相对于 $\{A\}$ 之姿态: ${}^A_R = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$

转动 -3

□ Ex: {B}相对于{A}之姿态 ${}^A_B R = ?$



上视图



蓝虚线: World Frame {A}

红实线: Body Frame {B}

$${}^A \hat{X}_B = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} 0.866 \\ 0.5 \\ 0 \end{bmatrix}$$

$${}^A \hat{Y}_B = \begin{bmatrix} \hat{Y}_B \cdot \hat{X}_A \\ \hat{Y}_B \cdot \hat{Y}_A \\ \hat{Y}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.866 \\ 0 \end{bmatrix}$$

$${}^A \hat{Z}_B = \begin{bmatrix} \hat{Z}_B \cdot \hat{X}_A \\ \hat{Z}_B \cdot \hat{Y}_A \\ \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

{B}相对于{A}之姿态:

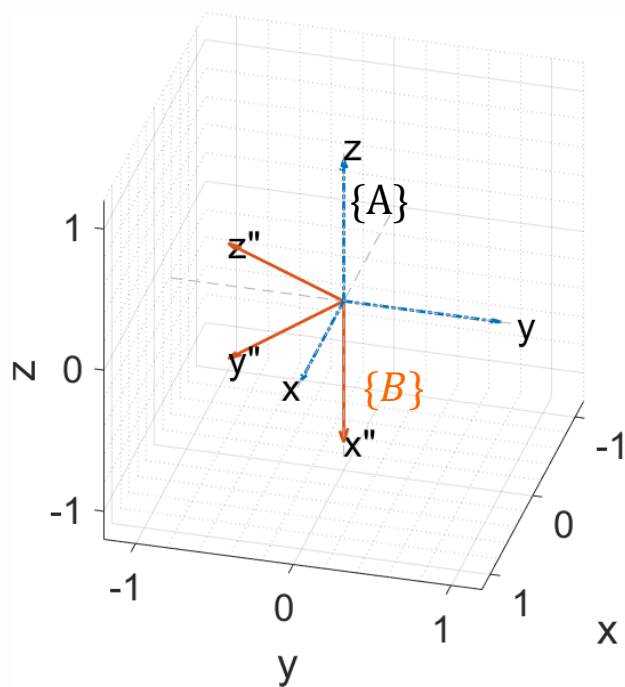
$${}^A_B R = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

转动 -4

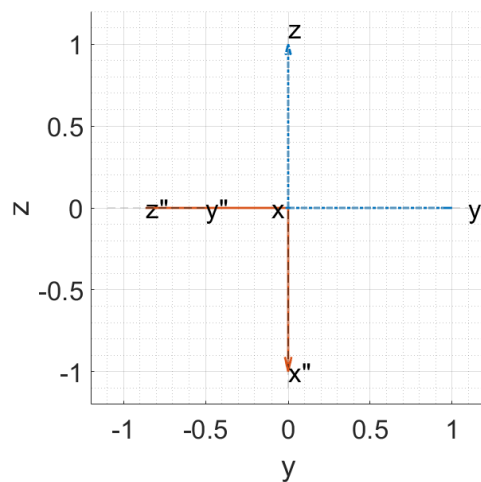
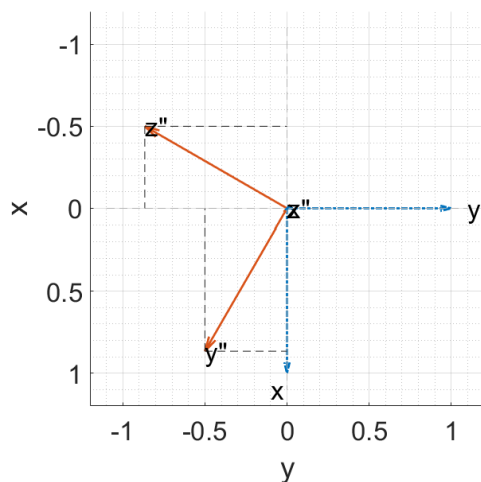
□ 课堂练习: $\{B\}$ 相对于 $\{A\}$ 的姿态 ${}^A_B R = ?$

蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$



投影到
XY/YZ平面

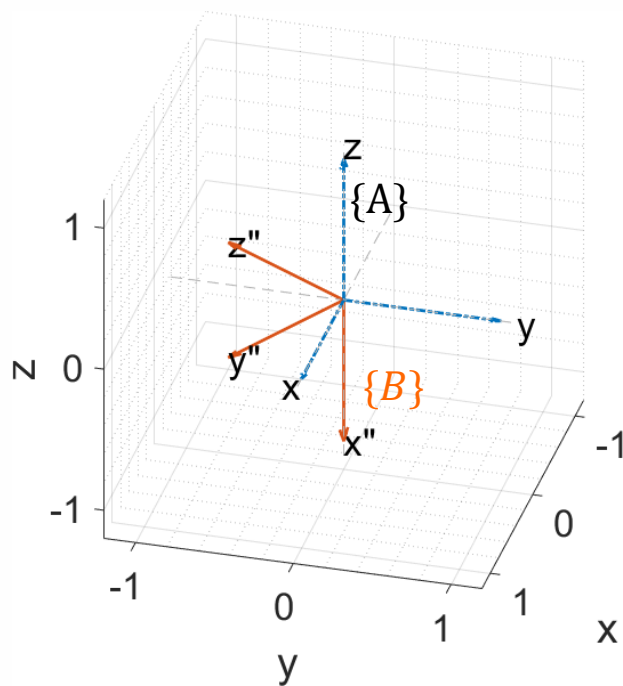


转动 -4

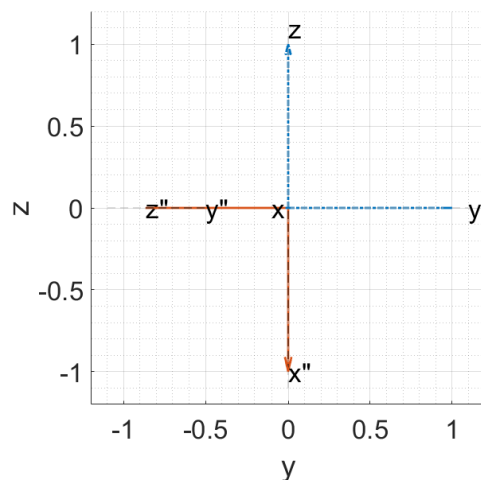
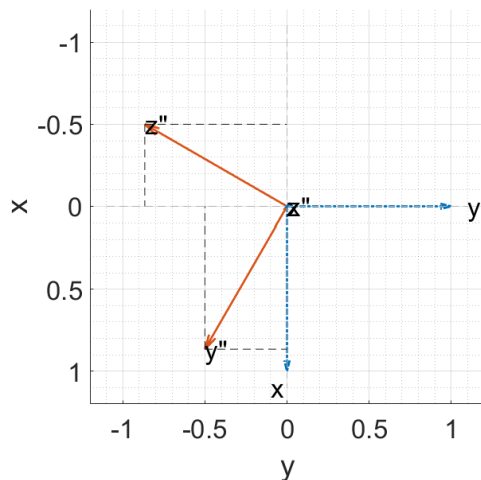
□ 课堂练习: $\{B\}$ 相对于 $\{A\}$ 的姿态 ${}^A_B R = ?$

蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$



投影到
XY/YZ平面



$${}^A\hat{X}_B = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$

$${}^A\hat{Y}_B = \begin{bmatrix} \hat{Y}_B \cdot \hat{X}_A \\ \hat{Y}_B \cdot \hat{Y}_A \\ \hat{Y}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} 0.866 \\ -0.5 \\ 0 \end{bmatrix}$$

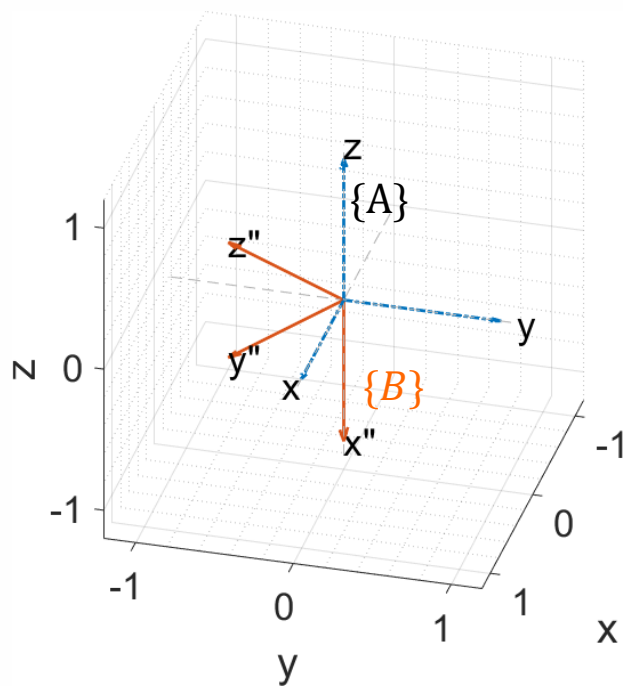
$${}^A\hat{Z}_B = \begin{bmatrix} \hat{Z}_B \cdot \hat{X}_A \\ \hat{Z}_B \cdot \hat{Y}_A \\ \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} -0.5 \\ -0.866 \\ 0 \end{bmatrix}$$

转动 -4

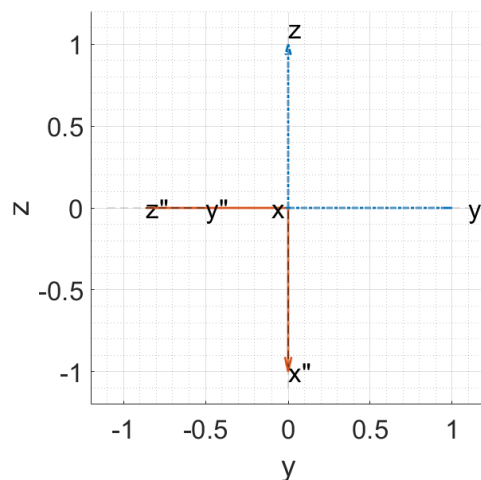
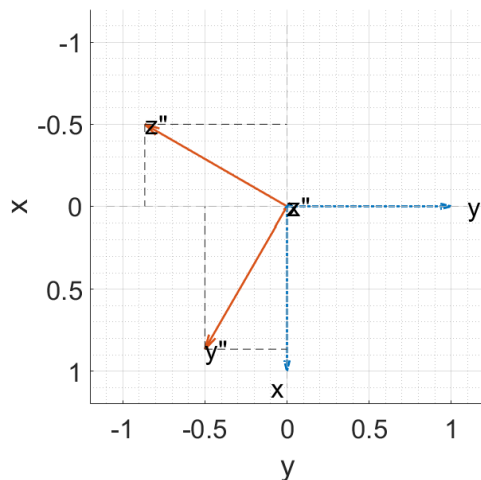
□ 课题联系: $\{B\}$ 相对于 $\{A\}$ 之姿态 ${}^A R \stackrel{?}{=} {}_B^A$

蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$



投影到
XY/YZ平面



A.
$$\begin{bmatrix} 0 & -0.866 & 0.5 \\ 0 & 0.5 & 0.866 \\ -1 & 0 & 0 \end{bmatrix}$$

B.
$$\begin{bmatrix} 0 & 0.866 & -0.5 \\ 0 & -0.5 & -0.866 \\ -1 & 0 & 0 \end{bmatrix}$$

C.
$$\begin{bmatrix} 0 & 0 & -1 \\ 0.866 & -0.5 & 0 \\ -0.5 & -0.866 & 0 \end{bmatrix}$$

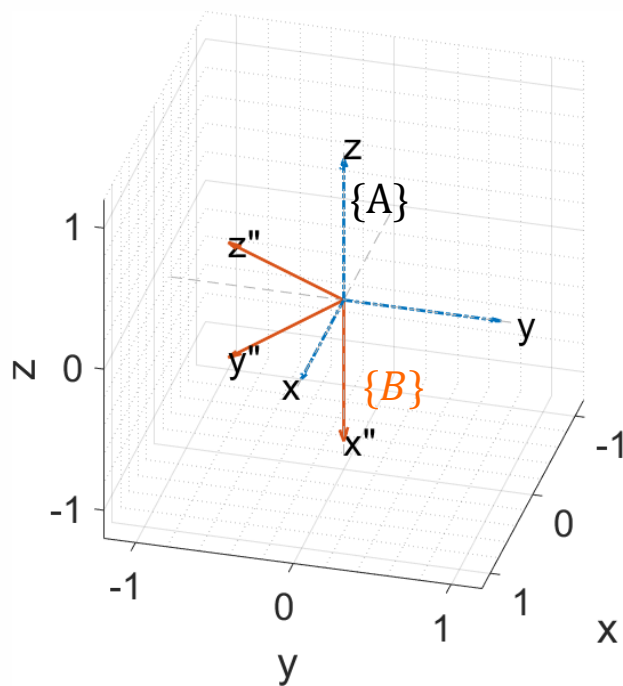
D.
$$\begin{bmatrix} 0 & 0 & -1 \\ -0.866 & 0.5 & 0 \\ 0.5 & 0.866 & 0 \end{bmatrix}$$

转动 -4

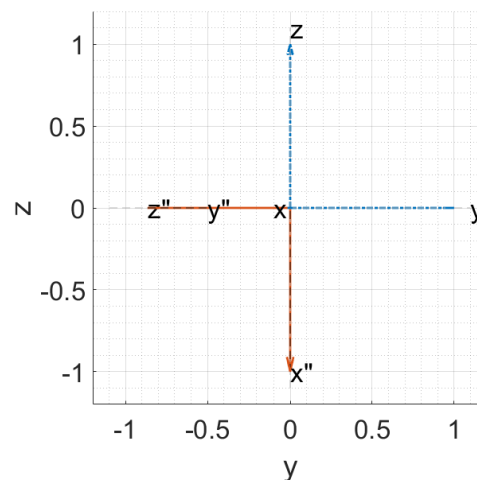
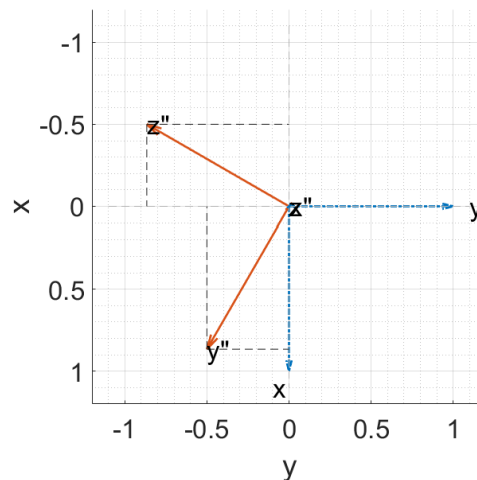
□ In-video Quiz: $\{B\}$ 相对于 $\{A\}$ 之姿态 ${}^A_R = ?$

蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$



投影到
XY/YZ平面



A.
$$\begin{bmatrix} 0 & -0.866 & 0.5 \\ 0 & 0.5 & 0.866 \\ -1 & 0 & 0 \end{bmatrix}$$

B.
$$\begin{bmatrix} 0 & 0.866 & -0.5 \\ 0 & -0.5 & -0.866 \\ -1 & 0 & 0 \end{bmatrix}$$

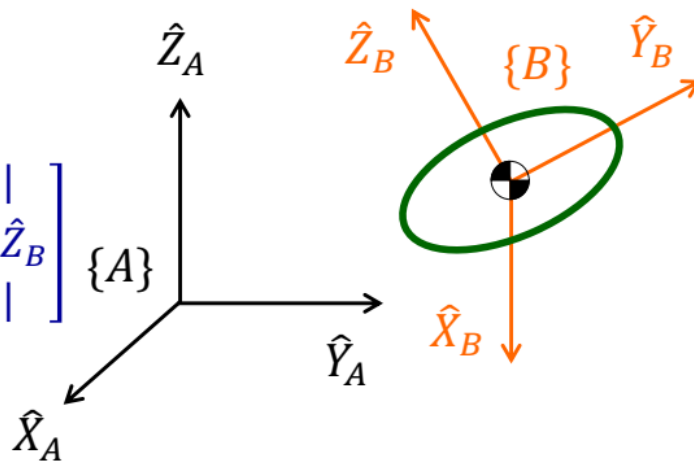
C.
$$\begin{bmatrix} 0 & 0 & -1 \\ 0.866 & -0.5 & 0 \\ -0.5 & -0.866 & 0 \end{bmatrix}$$

D.
$$\begin{bmatrix} 0 & 0 & -1 \\ -0.866 & 0.5 & 0 \\ 0.5 & 0.866 & 0 \end{bmatrix}$$

旋转矩阵 -1

□ 特性一

$${}^A_B R = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix} \{A\}$$



前后向量互换

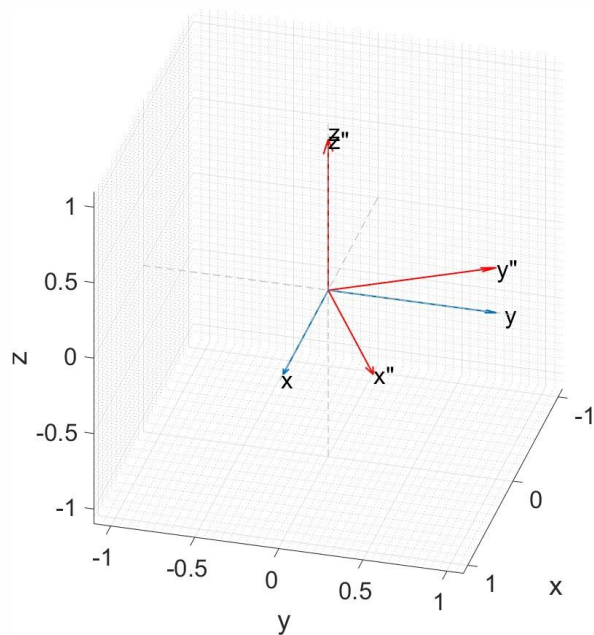
$$\begin{aligned} &= \begin{bmatrix} \hat{X}_A \cdot \hat{X}_B & \hat{X}_A \cdot \hat{Y}_B & \hat{X}_A \cdot \hat{Z}_B \\ \hat{Y}_A \cdot \hat{X}_B & \hat{Y}_A \cdot \hat{Y}_B & \hat{Y}_A \cdot \hat{Z}_B \\ \hat{Z}_A \cdot \hat{X}_B & \hat{Z}_A \cdot \hat{Y}_B & \hat{Z}_A \cdot \hat{Z}_B \end{bmatrix} = \begin{bmatrix} - & {}^B\hat{X}_A^T & - \\ - & {}^B\hat{Y}_A^T & - \\ - & {}^B\hat{Z}_A^T & - \end{bmatrix} \\ &= \begin{bmatrix} | & | & | \\ {}^B\hat{X}_A & {}^B\hat{Y}_A & {}^B\hat{Z}_A \\ | & | & | \end{bmatrix}^T = {}^B_A R^T \end{aligned}$$

$$\Rightarrow {}^A_B R = {}^B_A R^T$$

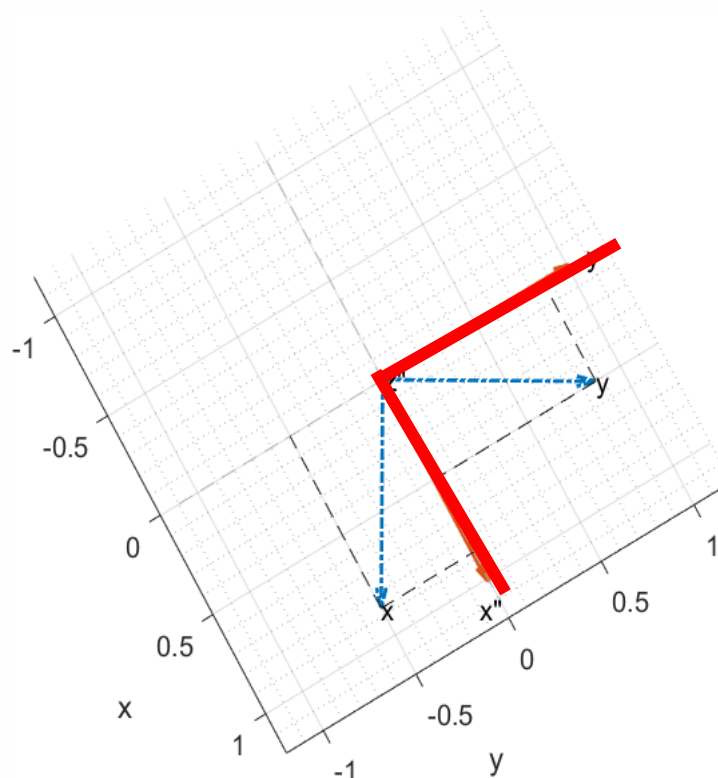
旋转矩阵 -2

□ Ex: $\{A\}$ 相对于 $\{B\}$ 之姿态 ${}^B_A R = ?$

以 $\{B\}$ 为参考基准



俯视图

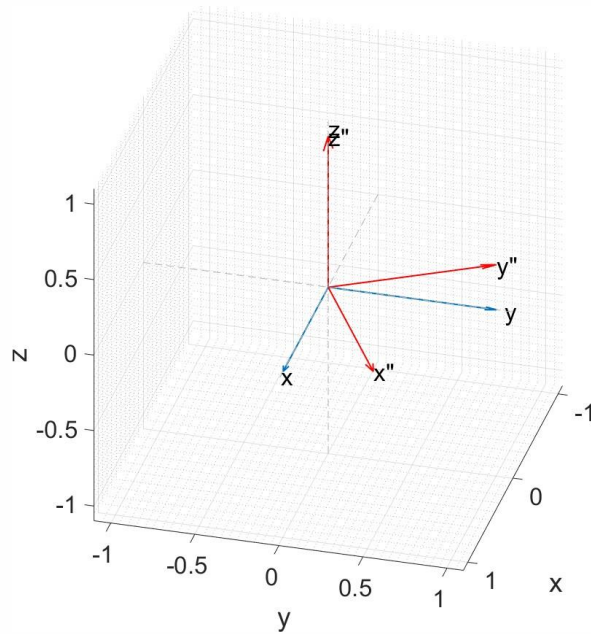


蓝虚线: World Frame $\{A\}$

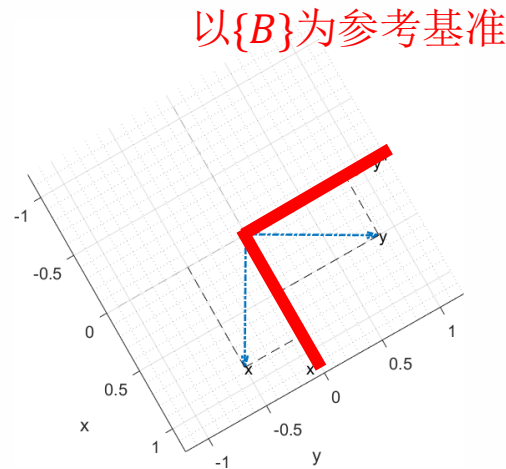
红实线: Body Frame $\{B\}$

旋转矩阵 -2

□ Ex: {A}相对于{B}之姿态 ${}^B_A R = ?$



俯视图



以{B}为参考基准

蓝虚线: World Frame {A}

红实线: Body Frame {B}

$${}^B\hat{X}_A = \begin{bmatrix} \hat{X}_A \cdot \hat{X}_B \\ \hat{Y}_A \cdot \hat{X}_B \\ \hat{Z}_A \cdot \hat{X}_B \end{bmatrix} = \begin{bmatrix} 0.866 \\ -0.5 \\ 0 \end{bmatrix}$$

$${}^B\hat{Y}_A = \begin{bmatrix} \hat{X}_A \cdot \hat{Y}_B \\ \hat{Y}_A \cdot \hat{Y}_B \\ \hat{Z}_A \cdot \hat{Y}_B \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.866 \\ 0 \end{bmatrix}$$

$${}^B\hat{Z}_A = \begin{bmatrix} \hat{X}_A \cdot \hat{Z}_B \\ \hat{Y}_A \cdot \hat{Z}_B \\ \hat{Z}_A \cdot \hat{Z}_B \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

{A}相对于{B}之姿态:

$${}^B_A R = \begin{bmatrix} 0.866 & 0.5 & 0 \\ -0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} = {}^A_B R^T$$

${}^A_B R$, “转动 -3 “页面结果

旋转矩阵 -3

□ 特性二

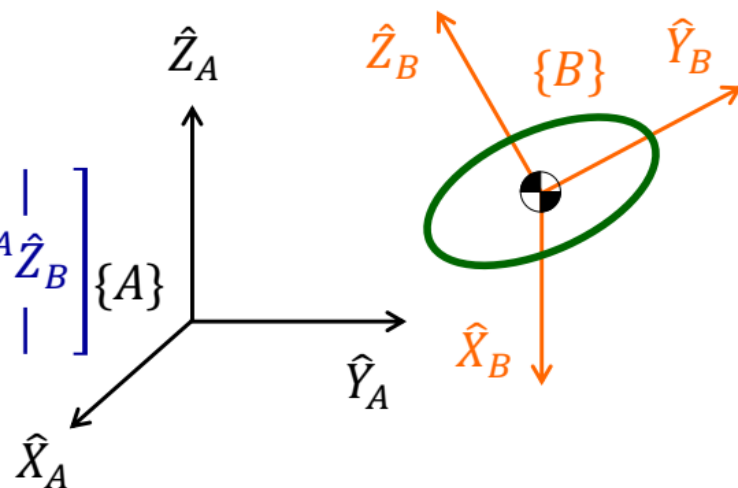
$${}^A_B R^T {}^A_B R = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix}^T \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix} \{A\}$$

$$= \begin{bmatrix} - & {}^A\hat{X}_B^T & - \\ - & {}^A\hat{Y}_B^T & - \\ - & {}^A\hat{Z}_B^T & - \end{bmatrix} \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix}$$

$$= I_3$$

↖ 3x3 identity matrix

$$= {}^A_B R^{-1} {}^A_B R$$



$$\Rightarrow {}^A_B R^T = {}^A_B R^{-1} = {}^B_A R$$

旋转矩阵 -4

□ A 3x3 正交矩阵 Q

$$QQ^T = Q^TQ = I$$

- ◆ 总是可逆的 $Q^{-1} = Q^T$
- ◆ 列：正交基
 - ◆ 长度=1
 - ◆ 相互垂直
- ◆ 旋转矩阵 (R)有9个数字，但上列两个条件置入了6个约束，所以R只有3个DOFs，与空间中转动具有3 DOFs相符
- ◆ Determinant =1 (rotation); =-1 (reflection)

旋转矩阵 -5 - 第二个功能

- 旋转矩阵除描述{B}相对于{A}之姿态，也可用于转换向量之坐标

original coordinate ${}^B P = {}^B P_x \hat{X}_B + {}^B P_y \hat{Y}_B + {}^B P_z \hat{Z}_B$

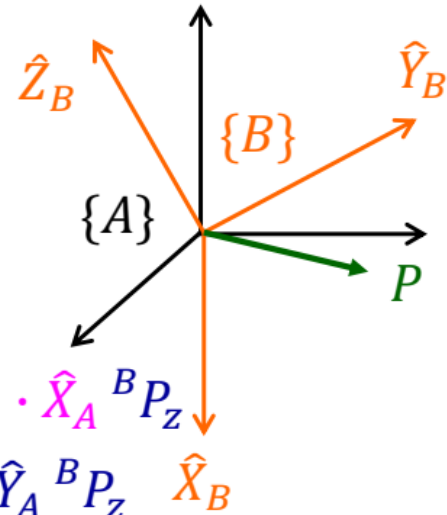
new coordinate ${}^A P = {}^A P_x \hat{X}_A + {}^A P_y \hat{Y}_A + {}^A P_z \hat{Z}_A$

where ${}^A P_x = {}^B P \cdot \hat{X}_A = \hat{X}_B \cdot \hat{X}_A {}^B P_x + \hat{Y}_B \cdot \hat{X}_A {}^B P_y + \hat{Z}_B \cdot \hat{X}_A {}^B P_z$

${}^A P_y = {}^B P \cdot \hat{Y}_A = \hat{X}_B \cdot \hat{Y}_A {}^B P_x + \hat{Y}_B \cdot \hat{Y}_A {}^B P_y + \hat{Z}_B \cdot \hat{Y}_A {}^B P_z$

${}^A P_z = {}^B P \cdot \hat{Z}_A = \hat{X}_B \cdot \hat{Z}_A {}^B P_x + \hat{Y}_B \cdot \hat{Z}_A {}^B P_y + \hat{Z}_B \cdot \hat{Z}_A {}^B P_z$

$$\Rightarrow {}^A P = \begin{bmatrix} {}^A P_x \\ {}^A P_y \\ {}^A P_z \end{bmatrix} = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} {}^B \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = {}^A_B R {}^B P$$

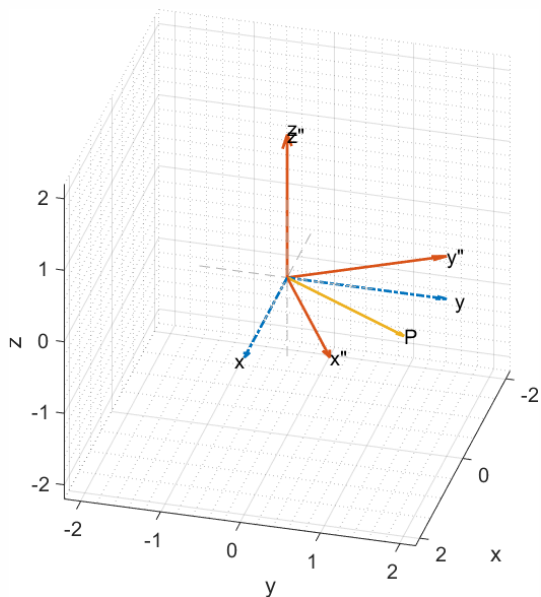


和“转动-1”页matrix相同，为旋转矩阵

旋转矩阵 -6

□ Ex: 若 $\{B\}$ 和 $\{A\}$ 的相对状态同“转动 -3”页面所示, 假设

$${}^B P = [1.732 \quad 1 \quad 0]^T, \quad {}^A P = ?$$



$${}^A_B R = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

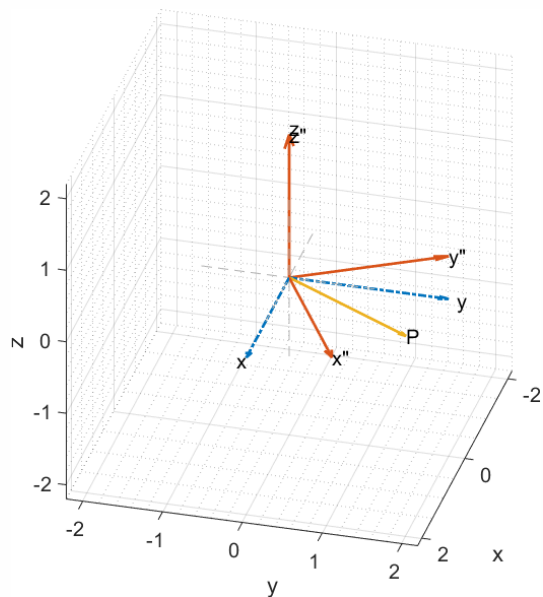
蓝虚线: World Frame $\{A\}$

红实线: Body Frame $\{B\}$

旋转矩阵 -6

□ Ex: 若 $\{B\}$ 和 $\{A\}$ 的相对状态同 “转动 -3 “页面所示, 假设

$${}^B P = [1.732 \quad 1 \quad 0]^T, \quad {}^A P = ?$$



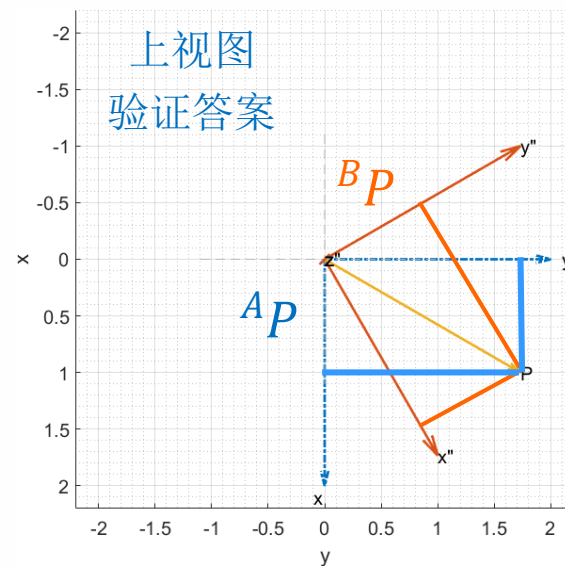
$${}^A P = {}^A_B R {}^B P$$

$${}^A P = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1.732 \\ 1 \\ 0 \end{bmatrix}$$

$${}^A P = \begin{bmatrix} 1 \\ 1.732 \\ 0 \end{bmatrix}$$

蓝虚线: World Frame $\{A\}$

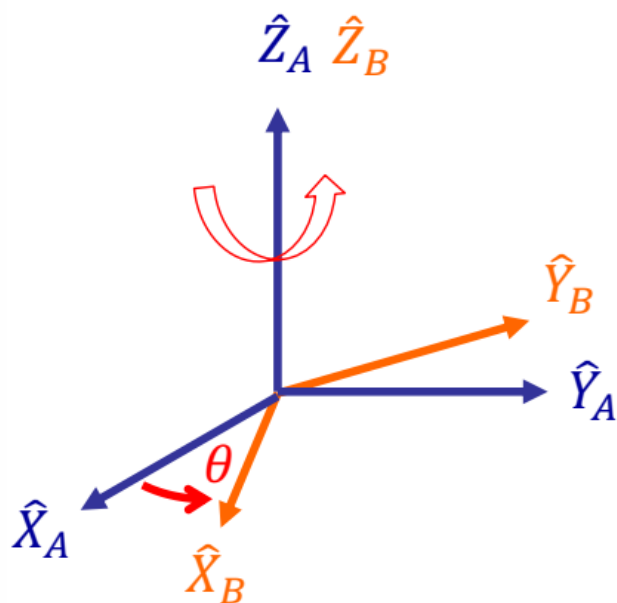
红实线: Body Frame $\{B\}$



旋转矩阵 -7

- 旋转矩阵的第三个功能，可进一步来描述物体转动的状态
- 以对三个主轴旋转的matrix为基础

- About $\hat{Z}_A$ with θ



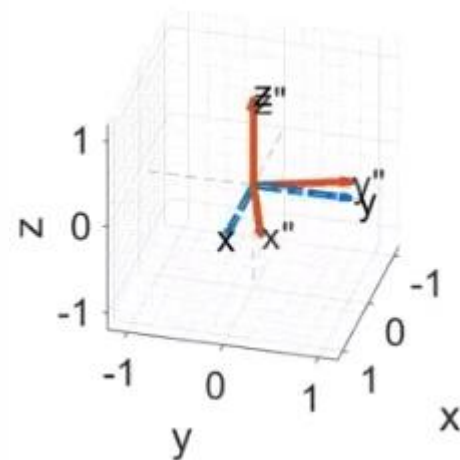
旋转角度

$$R_{\hat{Z}_A}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

旋转轴

$$= \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Note: ${}^A_B R = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix}$



旋转矩阵 -8

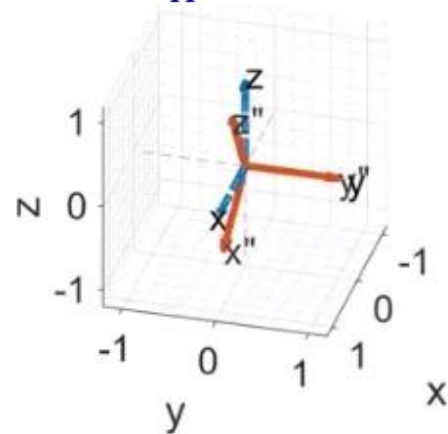
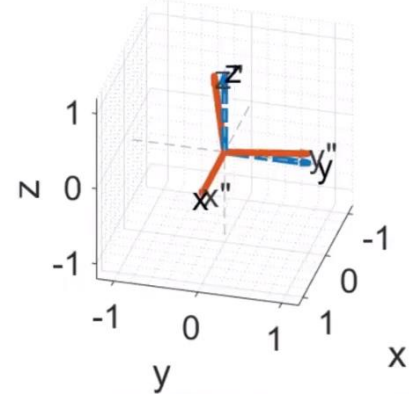
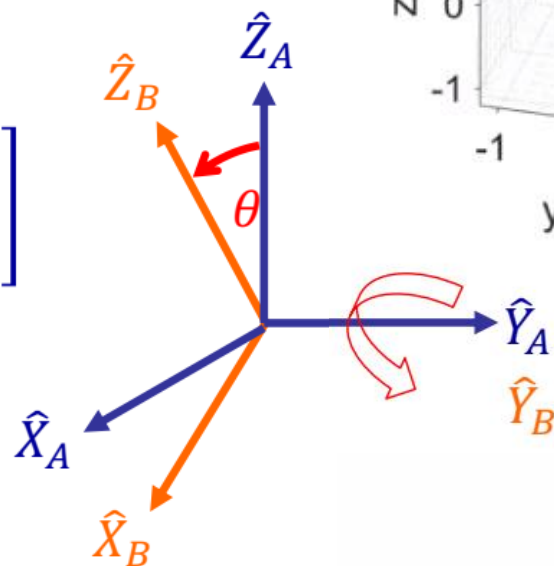
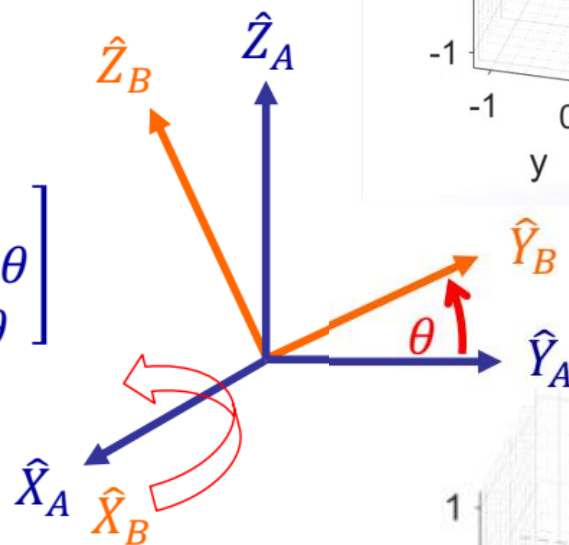
□ About $\hat{X}_A$ with θ

$$R_{\hat{X}_A}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & -s\theta \\ 0 & s\theta & c\theta \end{bmatrix}$$

□ About $\hat{Y}_A$ with θ

$$R_{\hat{Y}_A}(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} = \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}$$

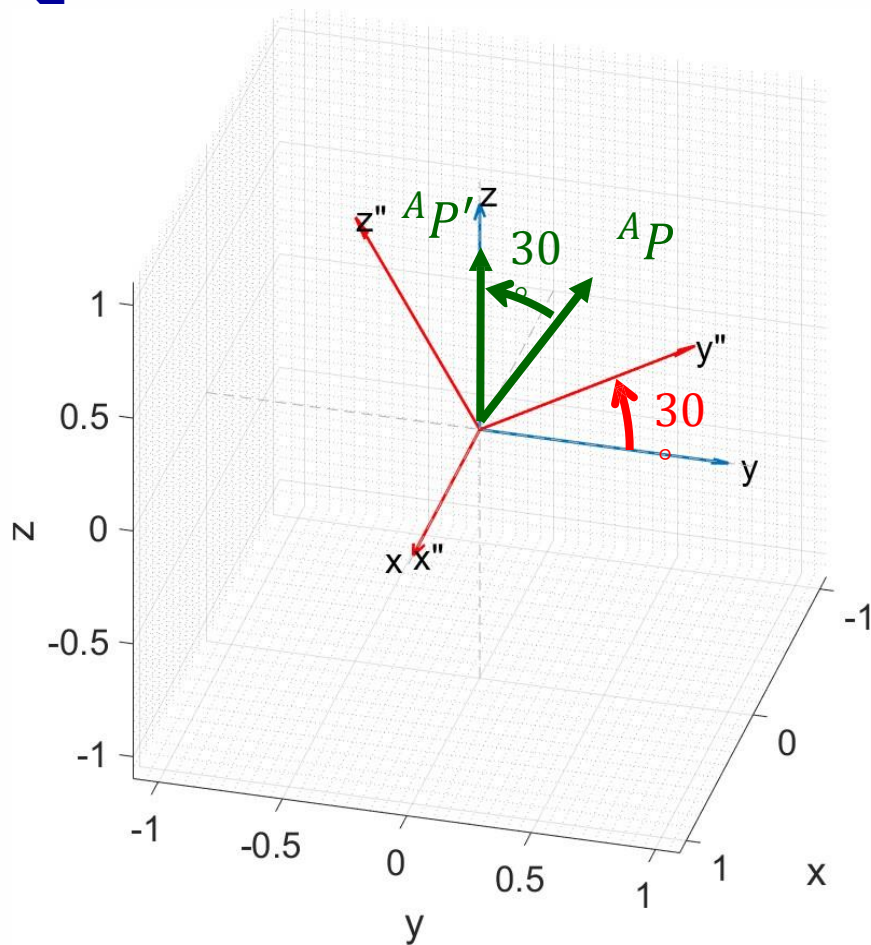
Note: ${}^A_B R = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix}$



旋转矩阵 -9

□ Ex: ${}^A P = [0 \quad 1 \quad 1.732]^T$ 对 $\hat{x}_A$ 轴旋转 30° , ${}^A P' = ?$

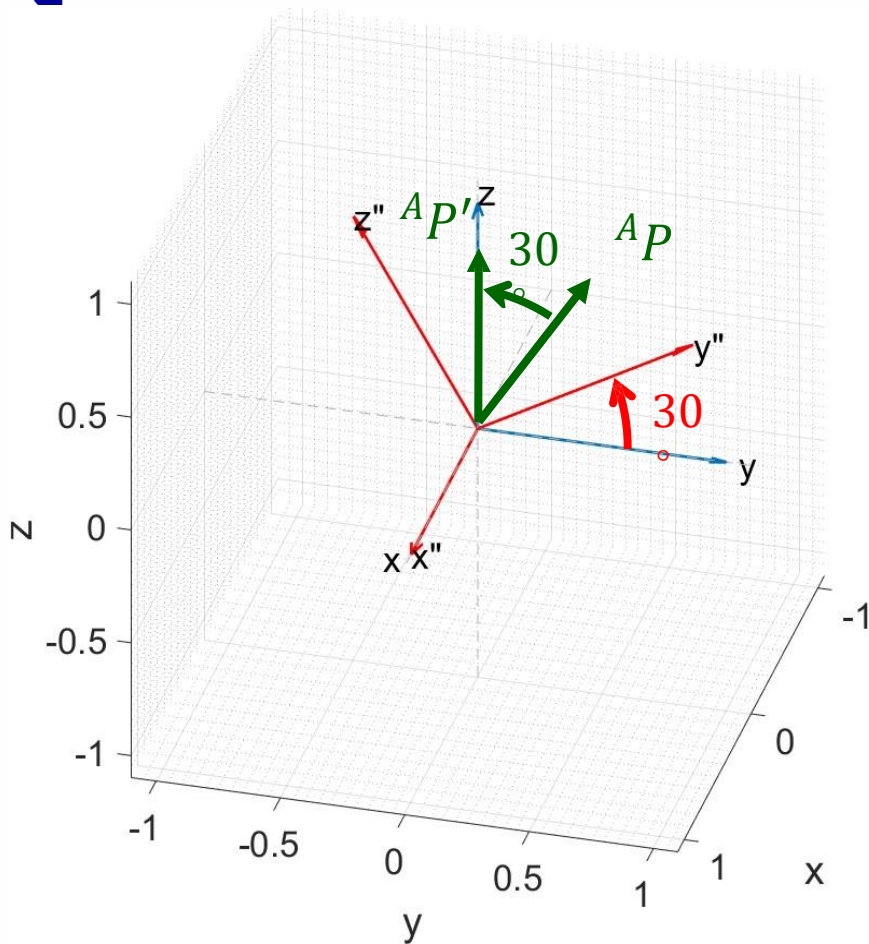
□



旋转矩阵 -9

□ Ex: ${}^A P = [0 \quad 1 \quad 1.732]^T$ 对 $\hat{x}_A$ 轴旋转 30° , ${}^A P'$ =?

□



$$R_{\hat{x}_A}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 30^\circ & -\sin 30^\circ \\ 0 & \sin 30^\circ & \cos 30^\circ \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & -0.5 \\ 0 & 0.5 & 0.866 \end{bmatrix}$$

$${}^A P' = R_{\hat{x}_A}(\theta) {}^A P$$

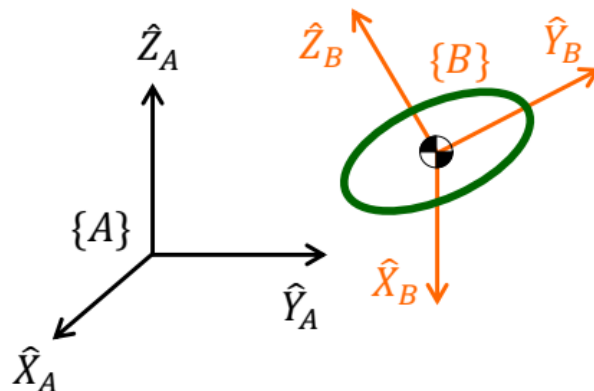
$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & -0.5 \\ 0 & 0.5 & 0.866 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1.732 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

旋转矩阵 -10

□ 旋转矩阵 的三种用法

1 描述一个frame(相对于另一个frame)的姿态

$${}^A_B R = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix}$$

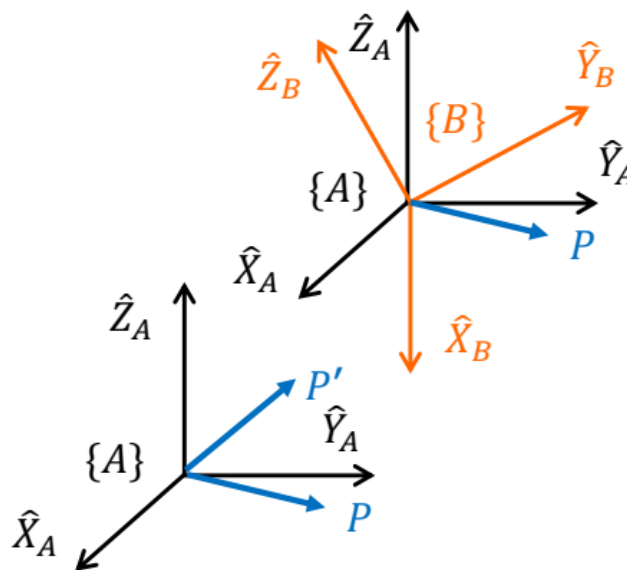


2 将point由某一个frame的表达换到另一个和此frame仅有相对转动的frame来表达

$${}^A P = {}^A_B R {}^B P$$

3 将point(vector)在同一个frame中进行转动

$${}^A P' = R(\theta) {}^A P$$

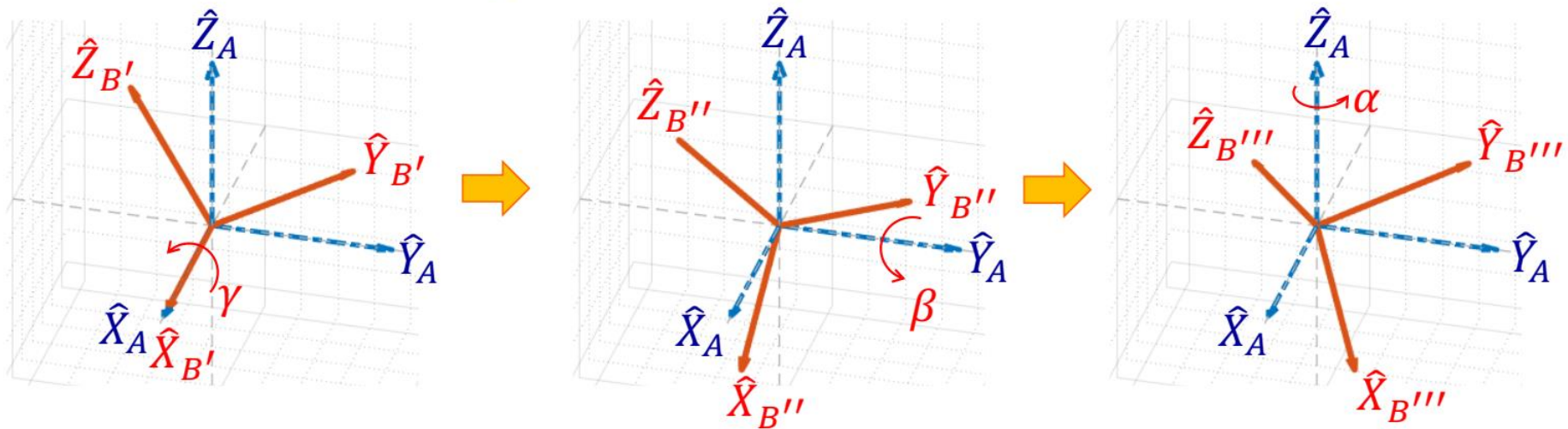


旋转矩阵与转角

- 空间中的Rotation是3 DOFs，那要如何把一般rotation matrix所表达的姿态，拆解成3次旋转角度，以对应到3个 DOFs?
- 拆解成“三次旋转连乘”所需注意事项
 - ◆ Rotation不是可交换的，所以多次旋转的先后顺序需要明确定义
 - ◆ 旋转转轴也需要明确定义。是对“固定不动”的转轴旋转？还是对“转动的frame当下所在”的转轴旋转？
- 两个拆解方式
 - ◆ 对方向“固定不动”的转轴旋转：固定角
 - ◆ 对“转动的frame当下所在”的转轴方向旋转：欧拉角

固定角 -1

□ X-Y-Z 固定角 – 由角度推算 R

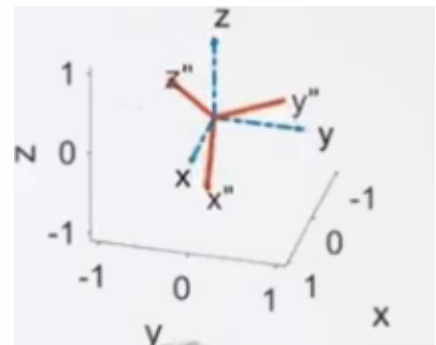


$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = R_Z(\alpha)R_Y(\beta)R_X(\gamma) \quad v' = {}^A_B Rv = R_3R_2R_1v$$

先转的放“后面”：以operator来想，对某一个向量，
“以同一个坐标为基准”，进行转动或移动的操作

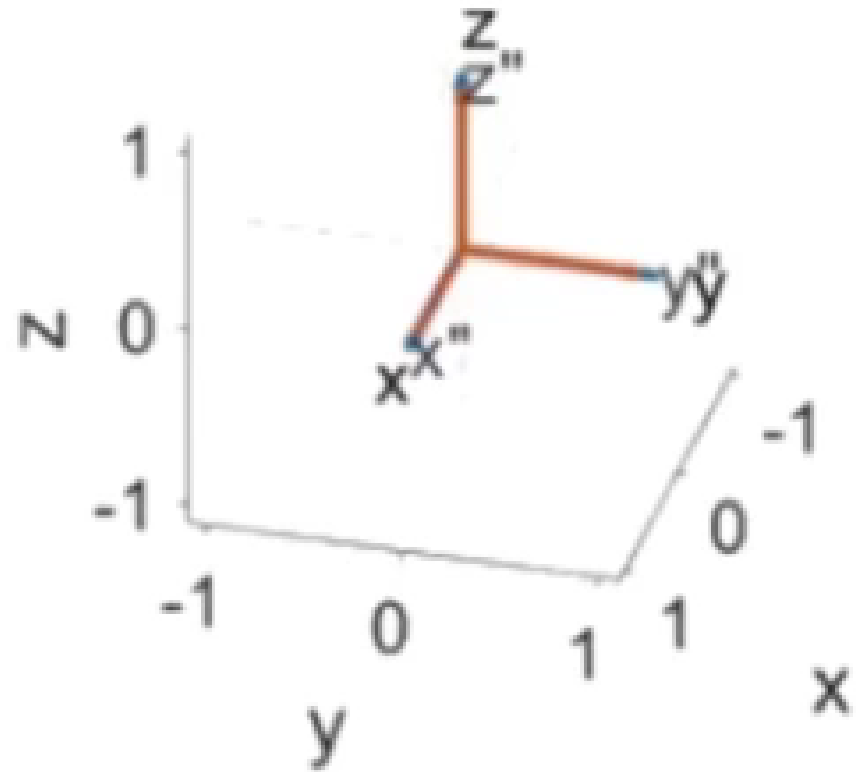
$$= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}$$

$$= \begin{bmatrix} cac\beta & cas\beta s\gamma - sac\gamma & cas\beta c\gamma + sas\gamma \\ sac\beta & sas\beta s\gamma + cac\gamma & sas\beta c\gamma - cas\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$



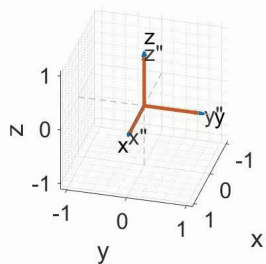
- Ex: 以固定角旋转：“先对X轴旋转60度，后对Y轴 旋转30度”和“先对Y轴旋转30度，后对X轴旋转60度”各自的 A_R_B 分别是？

先对X转60度，再对Y转30度



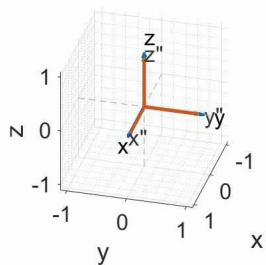
固定角 -2

- Ex: 以固定角旋转：“先对X轴旋转60度，后对Y轴 旋转30度”和“先对Y轴旋转30度，后对X轴旋转60度”各自的 A_R_B 分别是？



先对X转60度，再对Y转30度

$${}^A_R_B R_{XYZ}(\gamma, \beta, \alpha) = R_Z(0)R_Y(30)R_X(60) = \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix}$$

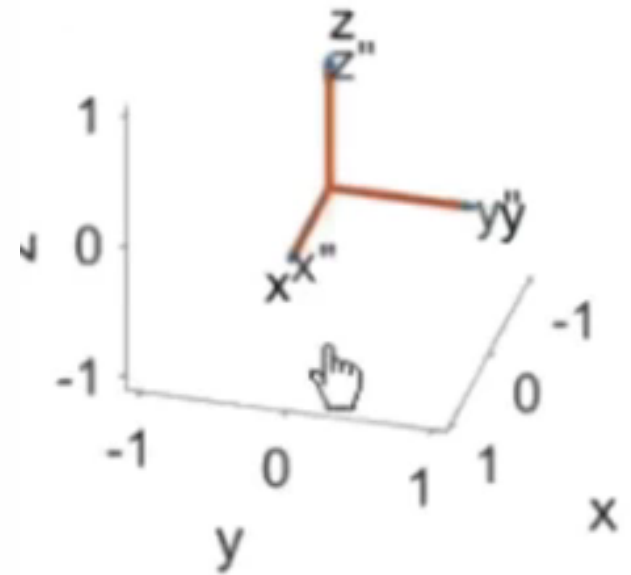


先对Y转30度，再对X转60度

$${}^A_R_B R_{XYZ}(\gamma, \beta, \alpha) = R_Z(0)R_X(60)R_Y(30) = \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0.433 & 0.5 & -0.75 \\ -0.25 & 0.866 & 0.433 \end{bmatrix}$$

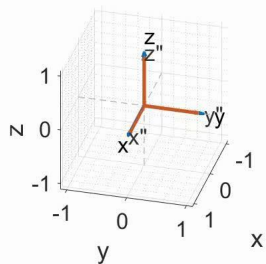
- Ex: 以固定角旋转：“先对X轴旋转60度，后对Y轴 旋转30度”和“先对Y轴旋转30度，后对X轴旋转60度”各自的 A_R_B 分别是？

再对Y转30度，先对X转60度，



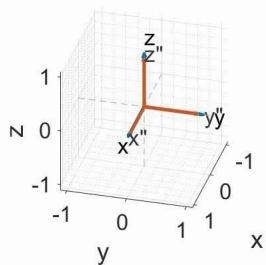
固定角 -2

- Ex: 以固定角旋转：“先对X轴旋转60度，后对Y轴 旋转30度”和“先对Y轴旋转30度，后对X轴旋转60度”各自的 A_R_B 分别是？



先对X转60度，再对Y转30度

$${}^A_R_B R_{XYZ}(\gamma, \beta, \alpha) = R_Z(0)R_Y(30)R_X(60) = \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix}$$



先对Y转30度，再对X转60度

$${}^A_R_B R_{XYZ}(\gamma, \beta, \alpha) = R_Z(0)R_X(60)R_Y(30) = \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0.433 & 0.5 & -0.75 \\ -0.25 & 0.866 & 0.433 \end{bmatrix}$$

固定角 -3

□ X-Y-Z 固定角 – 由 R 推算角度

$${}^A_B R_{XYZ}(\gamma, \beta, \alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

If $\beta \neq 90^\circ$

$$\beta = \text{Atan2}(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2})$$

$$\alpha = \text{Atan2}(r_{21}/c\beta, r_{11}/c\beta)$$

$$\gamma = \text{Atan2}(r_{32}/c\beta, r_{33}/c\beta)$$

$$-90^\circ \leq \beta \leq 90^\circ$$

Single solution

If $\beta = 90^\circ$

$$\alpha = 0^\circ$$

$$\gamma = \text{Atan2}(r_{12}, r_{22})$$

If $\beta = -90^\circ$

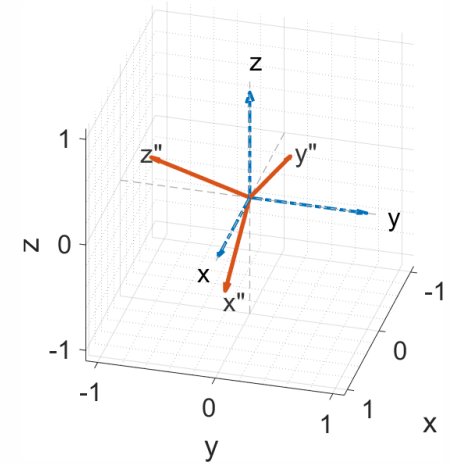
$$\alpha = 0^\circ$$

$$\gamma = -\text{Atan2}(r_{12}, r_{22})$$

固定角 -4

□ Ex: 以X-Y-Z 固定角方法, 反算 $R =$

$$\begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix} \text{的角度}$$



$$\begin{aligned} \beta &= \text{Atan2} \left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2} \right) = \text{Atan2} \left(-(-0.5), \sqrt{0.866^2 + 0^2} \right) = 30^\circ \\ \alpha &= \text{Atan2} \left(\frac{r_{21}}{c\beta}, \frac{r_{11}}{c\beta} \right) = \text{Atan2} \left(\frac{0}{\cos 30}, \frac{0.866}{\cos 30} \right) = 0^\circ \\ \gamma &= \text{Atan2} \left(\frac{r_{32}}{c\beta}, \frac{r_{33}}{c\beta} \right) = \text{Atan2} \left(\frac{0.75}{\cos 30}, \frac{0.433}{\cos 30} \right) = 60^\circ \end{aligned}$$



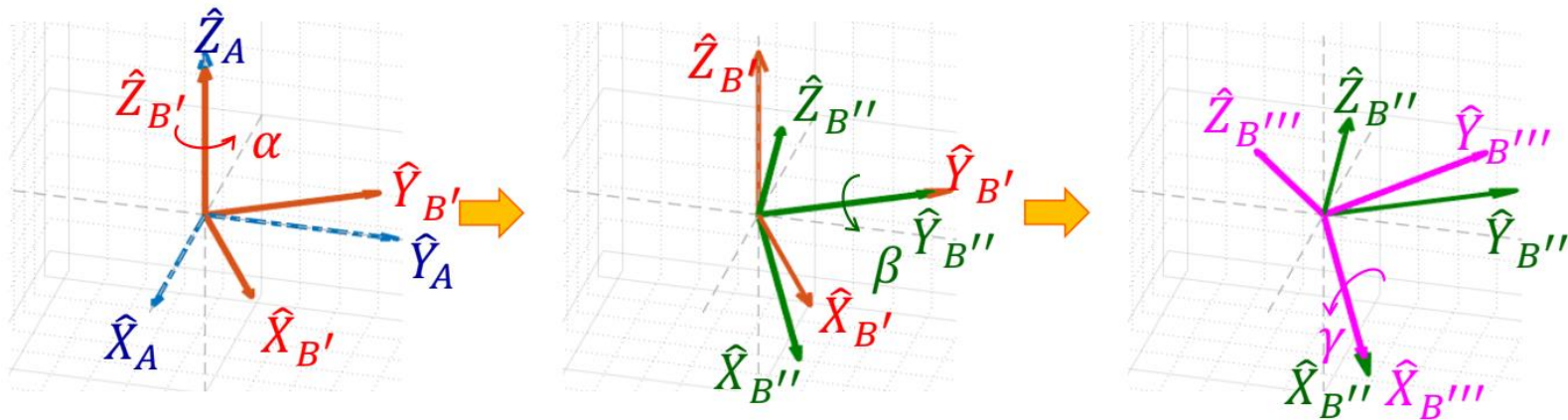
$R_Z(0)R_Y(30)R_X(60)$

先对X转60度, 再对Y转30度

和固定角 -2的结果相同

欧拉角 -1

□ Z-Y-X 欧拉角 - 由角度推算R



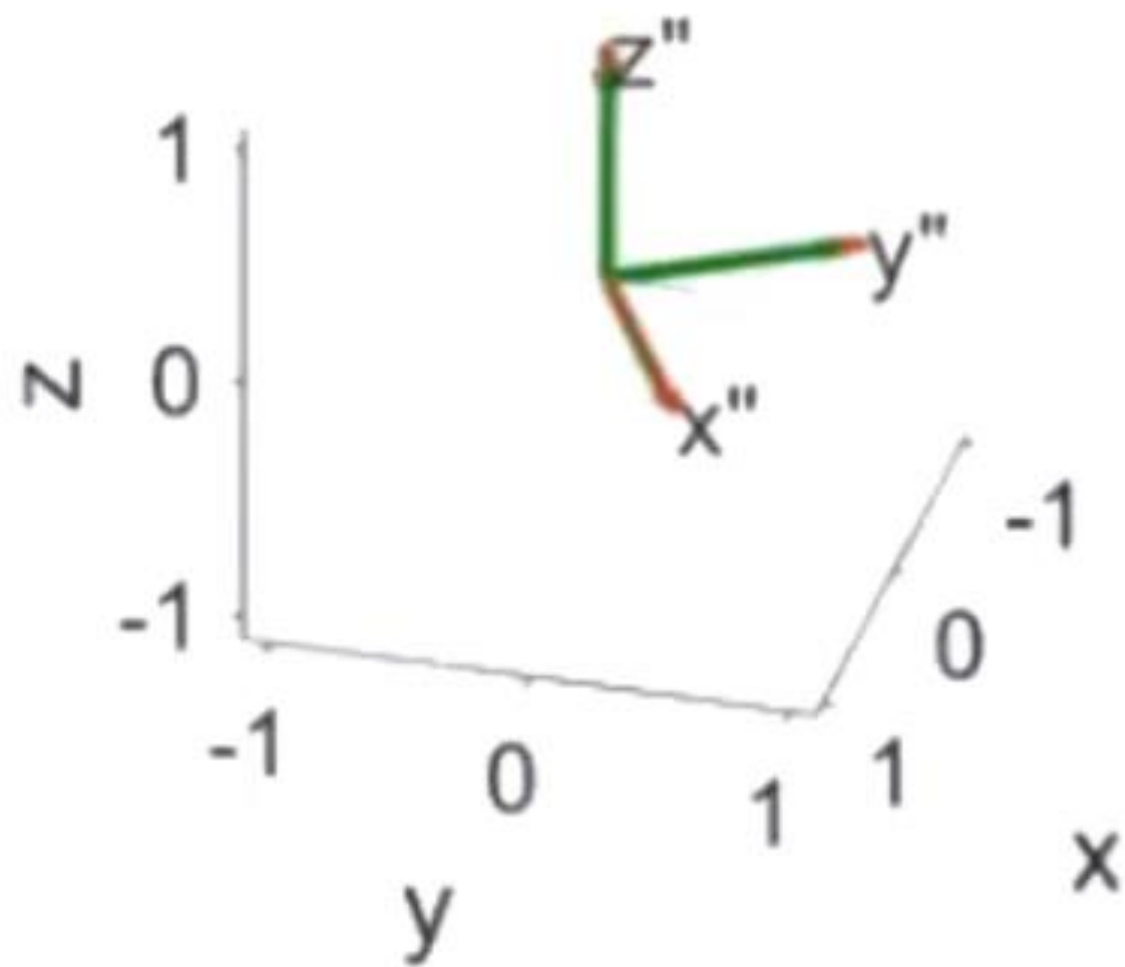
$${}^A_B R_{Z^F Y^F X^F}(\alpha, \beta, \gamma) = {}^{A^F}_{B^F} R_{{}^{B^F}_{B^{FF}}} R_{{}^{B^{FF}}_B} R = R_{Z^F}(\alpha) R_{Y^F}(\beta) R_{X^F}(\gamma)$$

先转的放“前面”：以mapping来想，对某一个向量，从最后一个frame“逐渐转动或移动”来回到第一个frame

$${}^A P = {}^A_B R {}^B P = R_1 R_2 R_3 {}^B P$$

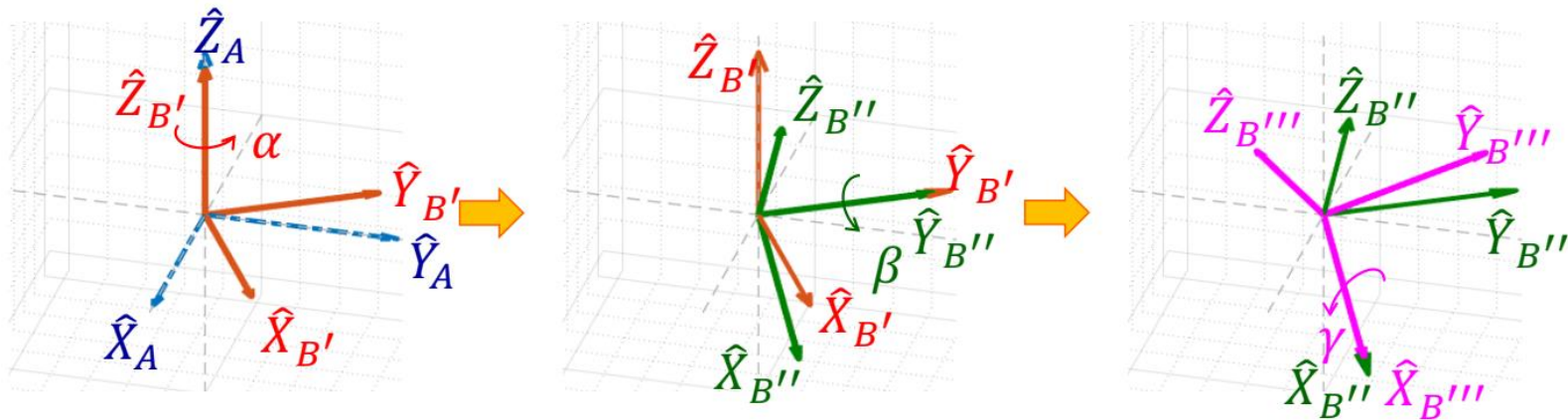
$$\begin{aligned} &= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix} \\ &= R_Z(\alpha) R_Y(\beta) R_X(\gamma) = {}^A_B R_{XYZ}(\gamma, \beta, \alpha) \end{aligned}$$

和X-Y-Z Fixed angle得到一样的R



欧拉角 -1

□ Z-Y-X 欧拉角 - 由角度推算R



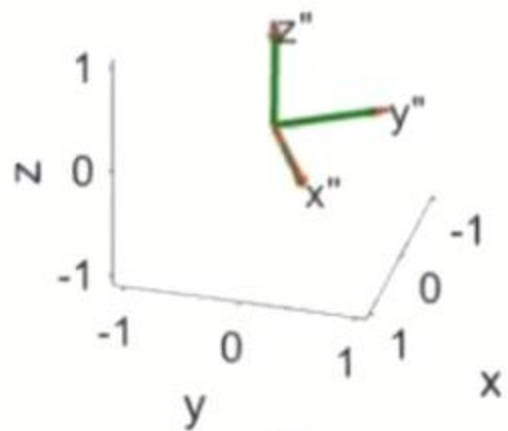
$${}^A_B R_{Z^F Y^F X^F}(\alpha, \beta, \gamma) = {}^{A^F}_{B^F} R_{{}^{B^F}_{B^{FF}}} R_{{}^{B^{FF}}_B} R = R_{Z^F}(\alpha) R_{Y^F}(\beta) R_{X^F}(\gamma)$$

先转的放“前面”：以mapping来想，对某一个向量，从最后一个frame“逐渐转动或移动”来回到第一个frame

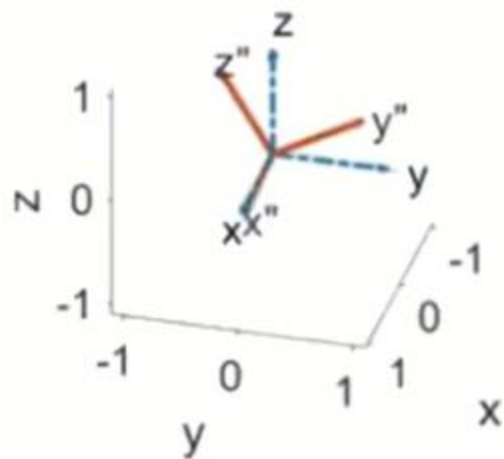
$${}^A P = {}^A_B R {}^B P = R_1 R_2 R_3 {}^B P$$

$$\begin{aligned} &= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix} \\ &= R_Z(\alpha) R_Y(\beta) R_X(\gamma) = {}^A_B R_{XYZ}(\gamma, \beta, \alpha) \end{aligned}$$

和X-Y-Z Fixed angle得到一样的R



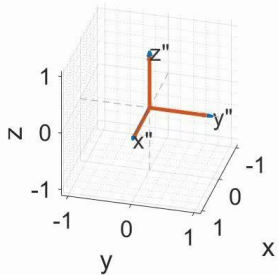
最后得出相同的 R



欧拉角 -2

- Ex: 以欧拉角旋转：“先对X轴旋转60度，后对Y轴 旋转30度”和“先对Y轴旋转30度，后对X轴旋转60度”各自的 A_R_B 分别是？

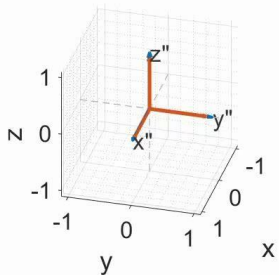
先对X转60度，再对Y转30度



$${}^A_R_B R_{X^F Y^F Z^F}(\gamma, \beta, \alpha) = R_{X^F}(60) R_{Y^F}(30) = \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0.433 & 0.5 & -0.75 \\ -0.25 & 0.866 & 0.433 \end{bmatrix}$$



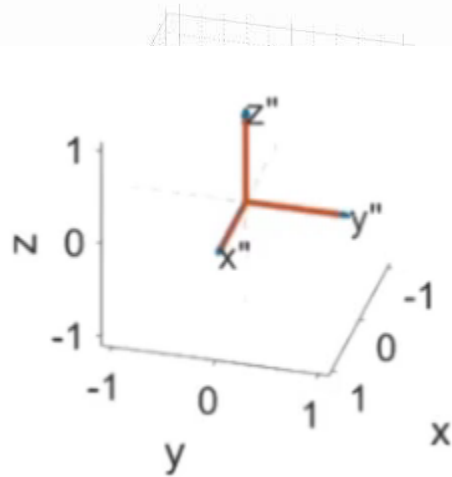
先对Y转30度，再对X转60度



$${}^A_R_B R_{X^F Y^F Z^F}(\gamma, \beta, \alpha) = R_{Y^F}(30) R_{X^F}(60) = \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix}$$

欧拉角 -3

- Ex: Euler(Y30, X60) v.s. Fixed(X60, Y30)



欧拉角:

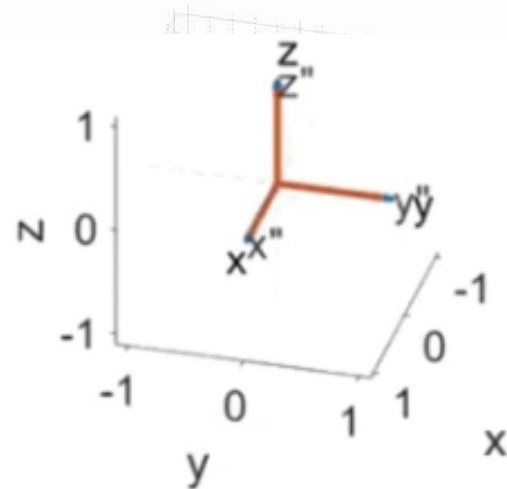
先对Y转30度, 再对X转60度

$${}^A_B R_{X'Y'Z'}(\gamma, \beta, \alpha)$$

$$= R_{Y'}(30)R_{X'}(60)$$

$$= \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix}$$

=



固定角:

先对X转60度, 再对Y转30度

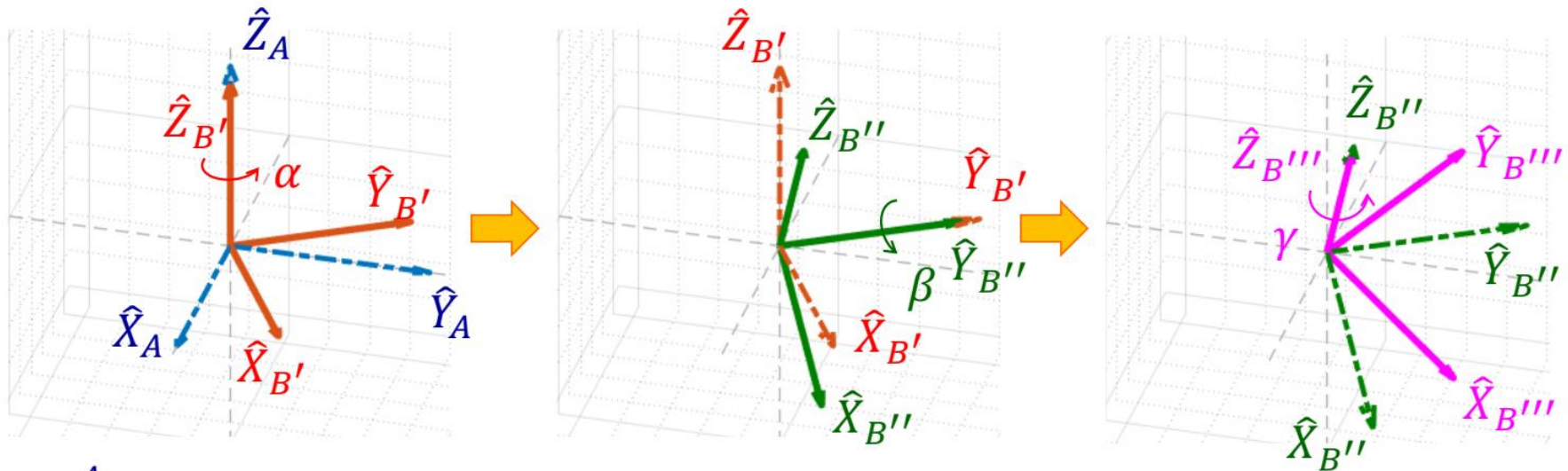
$${}^A_B R_{XYZ}(\gamma, \beta, \alpha)$$

$$= R_Y(30)R_X(60)$$

$$= \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix}$$

欧拉角 -4

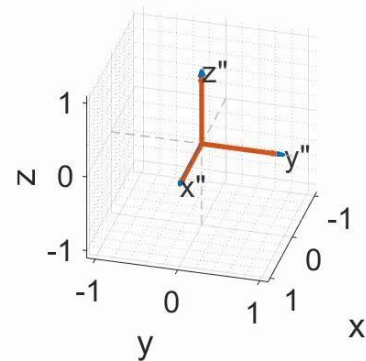
□ Z-Y-Z 欧拉角 - 由角度推算R



$${}^A_B R_{Z^F Y^F Z^F}(\alpha, \beta, \gamma) = R_{Z^F}(\alpha) R_{Y^F}(\beta) R_{Z^F}(\gamma)$$

先转的放“前面”

$$= \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix}$$



欧拉角 -5

□ Z-Y-Z 欧拉角 - 由R推算角度

$${}^A_B R_{Z'Y'Z'}(\alpha, \beta, \gamma) = \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}$$

If $\beta \neq 0^\circ$

$$\beta = \text{Atan2}(\sqrt{r_{31}^2 + r_{32}^2}, r_{33})$$

$$\alpha = \text{Atan2}(r_{23}/s\beta, r_{13}/s\beta)$$

$$\gamma = \text{Atan2}(r_{32}/s\beta, -r_{31}/s\beta)$$

If $\beta = 0^\circ$

$$\alpha = 0^\circ$$

$$\gamma = \text{Atan2}(-r_{12}, r_{11})$$

If $\beta = 180^\circ$

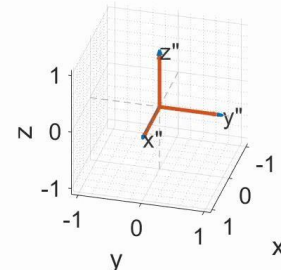
$$\alpha = 0^\circ$$

$$\gamma = \text{Atan2}(r_{12}, -r_{11})$$

欧拉角 -6

- Ex: Revisit 欧拉角-2的范例

$${}^A_B R_{X'Y'Z'}(60,30,0) = R_{X'}(60)R_{Y'}(30) = \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0.433 & 0.5 & -0.75 \\ -0.25 & 0.866 & 0.433 \end{bmatrix}$$



- 若以ZYZ的公式反算，欧拉角 为何？

$$R_{X^F}(60)R_{Y^F}(30)$$

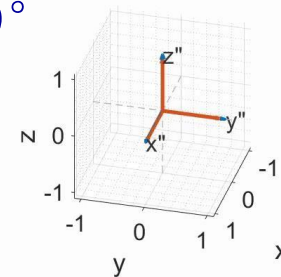
$$\beta = \text{Atan2}\left(\sqrt{r_{31}^2 + r_{32}^2}, r_{33}\right) = \text{Atan2}\left(\sqrt{(-0.25)^2 + 0.866^2}, 0.433\right) = 64.3^\circ$$

$$\alpha = \text{Atan2}\left(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}\right) = \text{Atan2}\left(\frac{-0.75}{s\beta}, \frac{0.5}{s\beta}\right) = -56.3^\circ$$

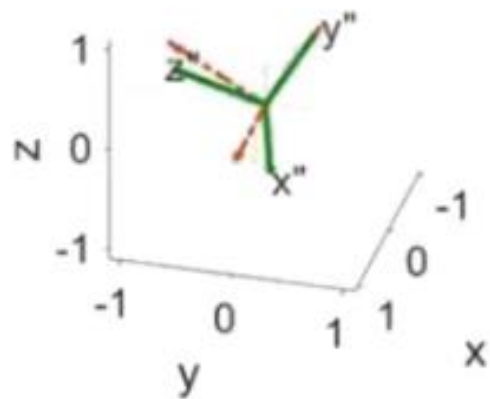
$$\gamma = \text{Atan2}(r_{32}/s\beta, -r_{31}/s\beta) = \text{Atan2}(0.866/s\beta, 0.25/s\beta) = 73.9^\circ$$

➡ $R_{Z'}(-56.3)R_{Y'}(64.3)R_{Z'}(73.9)$

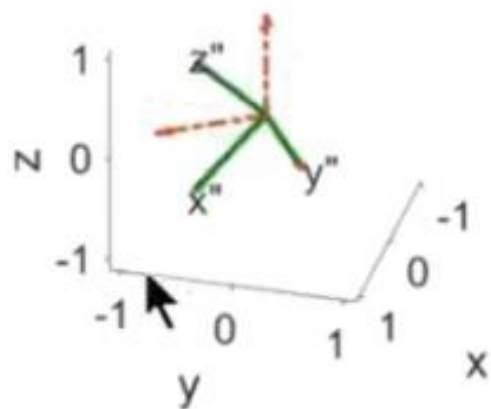
先对Z转 -56.3° , 对Y转 64.3° , 最后对Z转 73.9°



$$R_{Z^F}(-56.3)R_{Y^F}(64.3)R_{Z^F}(73.9)$$



3° $R_{X'}(60)R_{Y'}(30)$

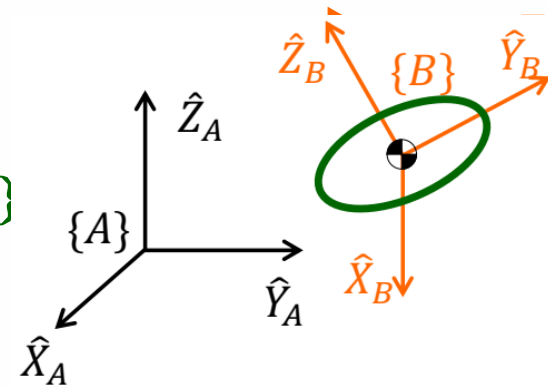


刚体状态的表达 -1

- “导读-3 “的问题：该如何整合表达刚体的状态？
- ⇒ 在刚体(Rigid body)上建立frame，常建立在质心上

- 移动：由body frame 的原点位置判定

$${}^A P_{B \text{ org}} = \begin{bmatrix} P_{B x} \\ P_{B y} \\ P_{B z} \end{bmatrix} = \text{origin of } \{B\} \text{ represented in } \{A\}$$



- 转动：由body frame的姿态判定

$${}^A R_B = \begin{bmatrix} \hat{X}_B \cdot \hat{X}_A & \hat{Y}_B \cdot \hat{X}_A & \hat{Z}_B \cdot \hat{X}_A \\ \hat{X}_B \cdot \hat{Y}_A & \hat{Y}_B \cdot \hat{Y}_A & \hat{Z}_B \cdot \hat{Y}_A \\ \hat{X}_B \cdot \hat{Z}_A & \hat{Y}_B \cdot \hat{Z}_A & \hat{Z}_B \cdot \hat{Z}_A \end{bmatrix} = \begin{bmatrix} | & | & | \\ {}^A \hat{X}_B & {}^A \hat{Y}_B & {}^A \hat{Z}_B \\ | & | & | \end{bmatrix}$$

- 汇整后：

$$\{B\} = \{ {}^A R_B, {}^A P_{B \text{ org}} \}$$

但无法进行量化计算

刚体状态的表达 -2

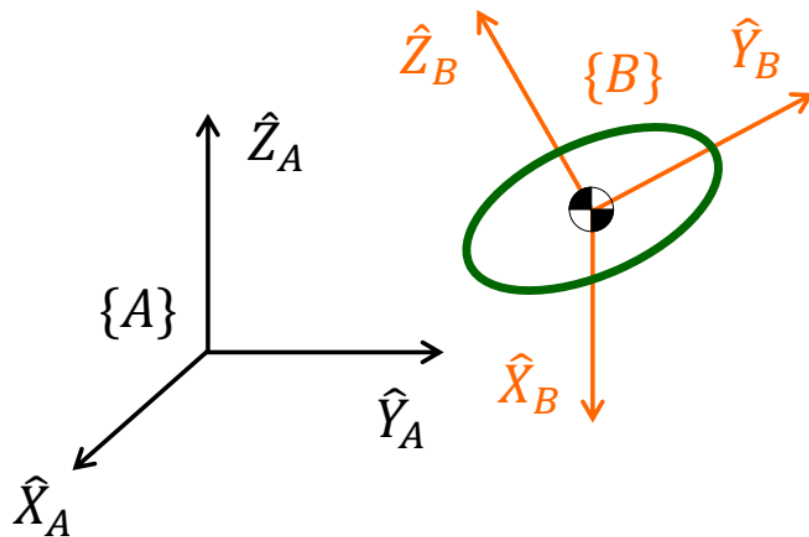
□ 如何将移动和转动整合在一起描述？

⇒ 齐次变换矩阵

$$\begin{bmatrix} {}^A_B R_{3 \times 3} & {}^A P_{B \text{ org } 3 \times 1} \\ \hline 0 & 1 \end{bmatrix}_{4 \times 4}$$

$$= \begin{bmatrix} | & | & | & | \\ {}^A \hat{X}_B & {}^A \hat{Y}_B & {}^A \hat{Z}_B & {}^A P_{B \text{ org }} \\ \hline | & | & | & | \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= {}^A_B T$$



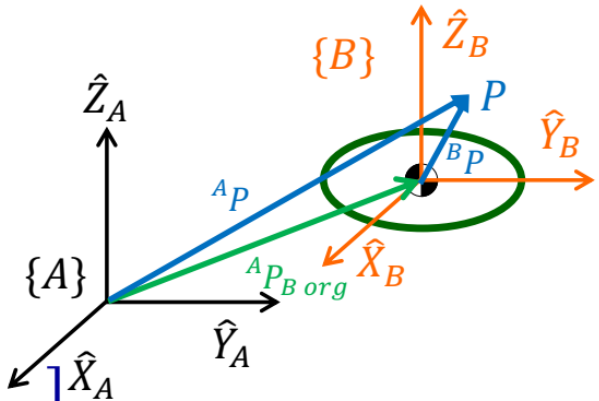
Mapping -1

- 以 Mapping, 转换向量 (或点) 之坐标系的方式来确认 ${}^A T$ 运算之正确性

◆ 仅有 移动

$${}^A P_{3 \times 1} = {}^B P_{3 \times 1} + {}^A P_{B \text{ org}}_{3 \times 1}$$

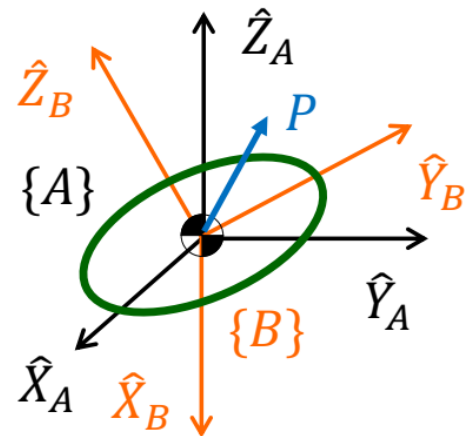
$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} I_{3 \times 3} & {}^A P_{B \text{ org}} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^B P + {}^A P_{B \text{ org}} \\ 1 \end{bmatrix}$$



◆ 仅有 转动

$${}^A P_{3 \times 1} = {}^A_B R_{3 \times 3} {}^B P_{3 \times 1}$$

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B R & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A_B R {}^B P \\ 1 \end{bmatrix}$$



Mapping -2

◆ 移动和转动复合

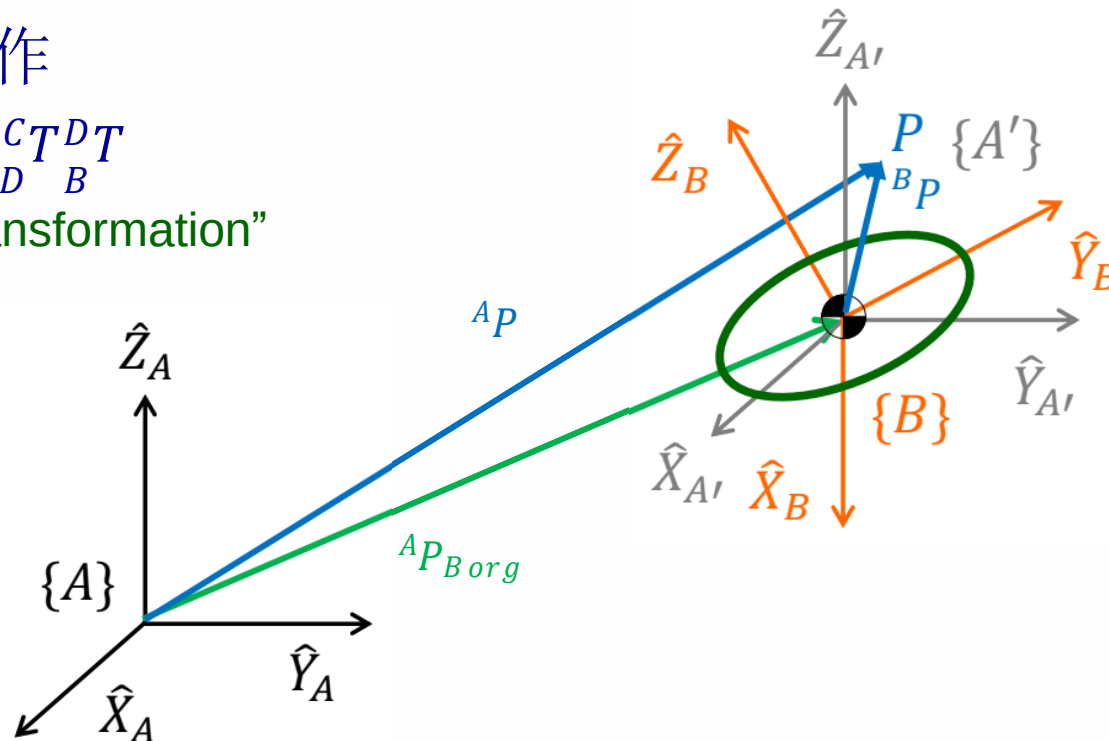
$${}^A P_{3 \times 1} = {}^A R {}^B P_{3 \times 1} + {}^A P_{B \text{ org } 3 \times 1}$$

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R & {}^A P_{B \text{ org }} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R {}^B P + {}^A P_{B \text{ org }} \\ 1 \end{bmatrix}$$

□ 可连续操作

$${}^A T_B = {}^A T_C {}^C T_D {}^D T_B$$

“sequential transformation”

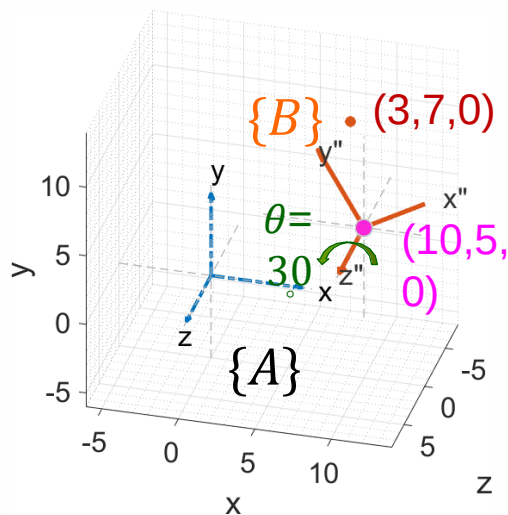


Mapping -3

□ Ex:

$${}^B P = \begin{bmatrix} 3 \\ 7 \\ 0 \end{bmatrix} \quad {}^A P_{Borg} = \begin{bmatrix} 10 \\ 5 \\ 0 \end{bmatrix} \quad {}^A \hat{X}_B = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix} \quad {}^A \hat{Y}_B = \begin{bmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \\ 2 \\ 0 \end{bmatrix} \quad {}^A \hat{Z}_B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow {}^A P = ?$$

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A R_B & {}^A P_{Borg} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B P \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 & 10 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 & 5 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{Transformation Matrix}} \begin{bmatrix} 3 \\ 7 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 9.098 \\ 12.562 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} | \\ {}^A P \\ | \\ 1 \end{bmatrix}$$

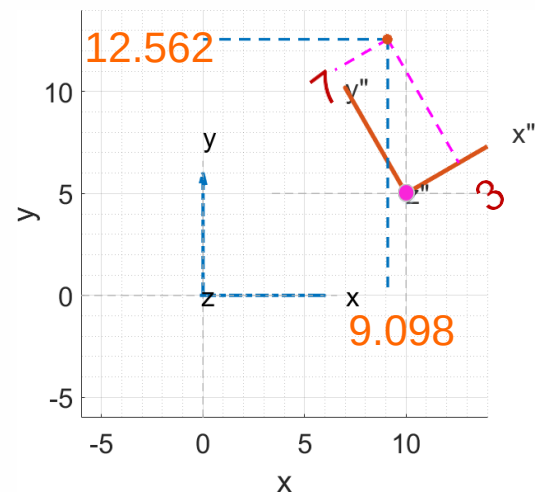


单纯看 T : 表达 $\{B\}$ 相对于 $\{A\}$ 的方法

看整个操作:

转换point在不同frame下的表达

投影至XY平面验证答案



Operators -1

□ ${}^A_B T$ 除了Mapping之外，也可当Operator，对向量（或点）进行移动或转动

◆ 仅有 移动

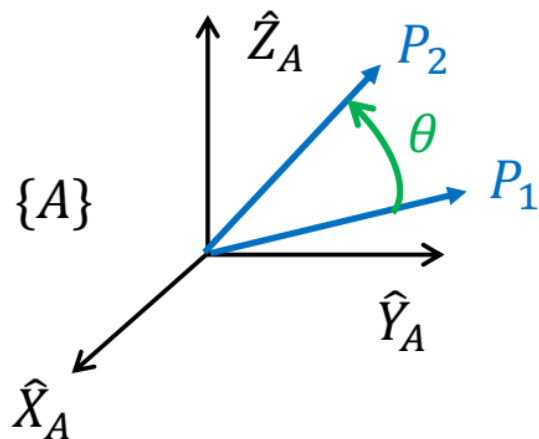
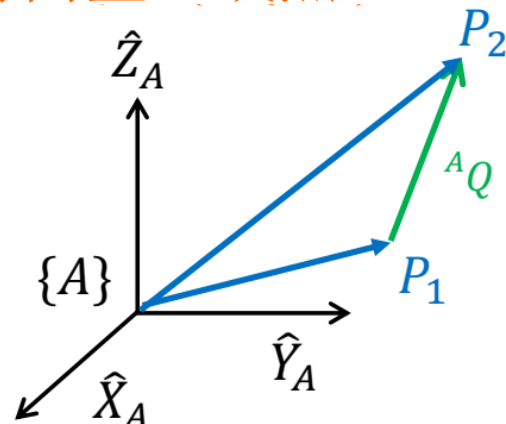
$${}^A P_{2 \times 1} = {}^A P_{1 \times 1} + {}^A Q_{\begin{smallmatrix} 3 \times \\ 1 \end{smallmatrix}}$$

$$\begin{bmatrix} {}^A P_2 \\ 1 \end{bmatrix} = D(Q) \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix} = \begin{bmatrix} I & {}^A Q \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix} = \begin{bmatrix} {}^A P_1 + {}^A Q \\ 1 \end{bmatrix}$$

◆ 仅有 转动

$${}^A P_{2 \times 1} = R_{\hat{K}}(\theta) {}^A P_{1 \times 1}$$

$$\begin{bmatrix} {}^A P_2 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\hat{K}}(\theta) & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\hat{K}}(\theta) {}^A P_1 \\ 1 \end{bmatrix}$$

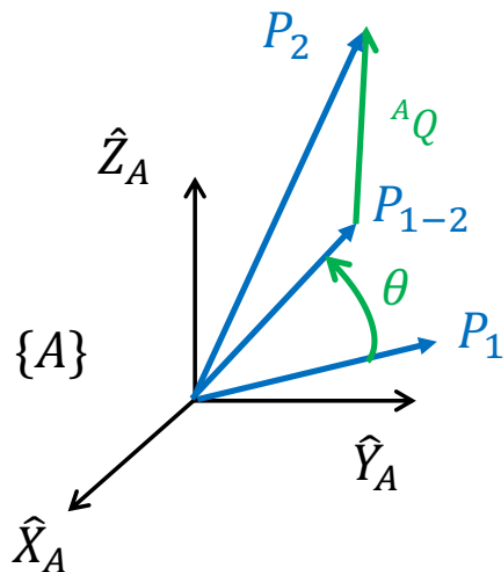


Operators -2

◆ 移动和转动复合

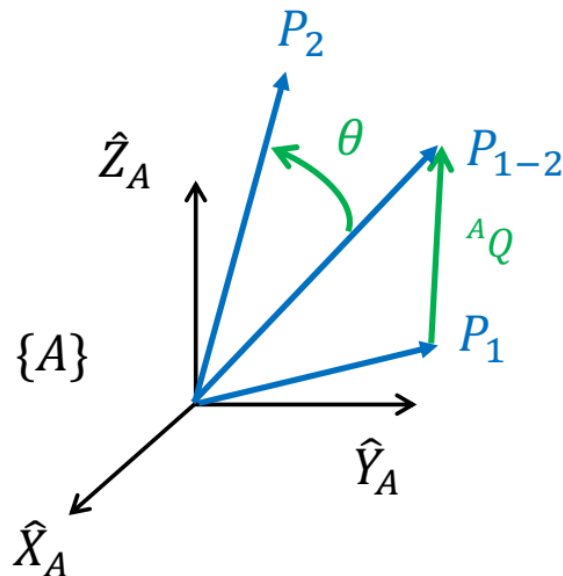
$${}^A P_2 = R_{\hat{K}}(\theta) {}^A P_1 + {}^A Q \quad \text{先转动再移动}$$

$$\begin{bmatrix} {}^A P_2 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\hat{K}}(\theta) & {}^A Q \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\hat{K}}(\theta) {}^A P_1 + {}^A Q \\ 1 \end{bmatrix} = T \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix}$$



先转动再移动

$\neq$



先移动再转动 (${}^A Q$ 也会被转动到)

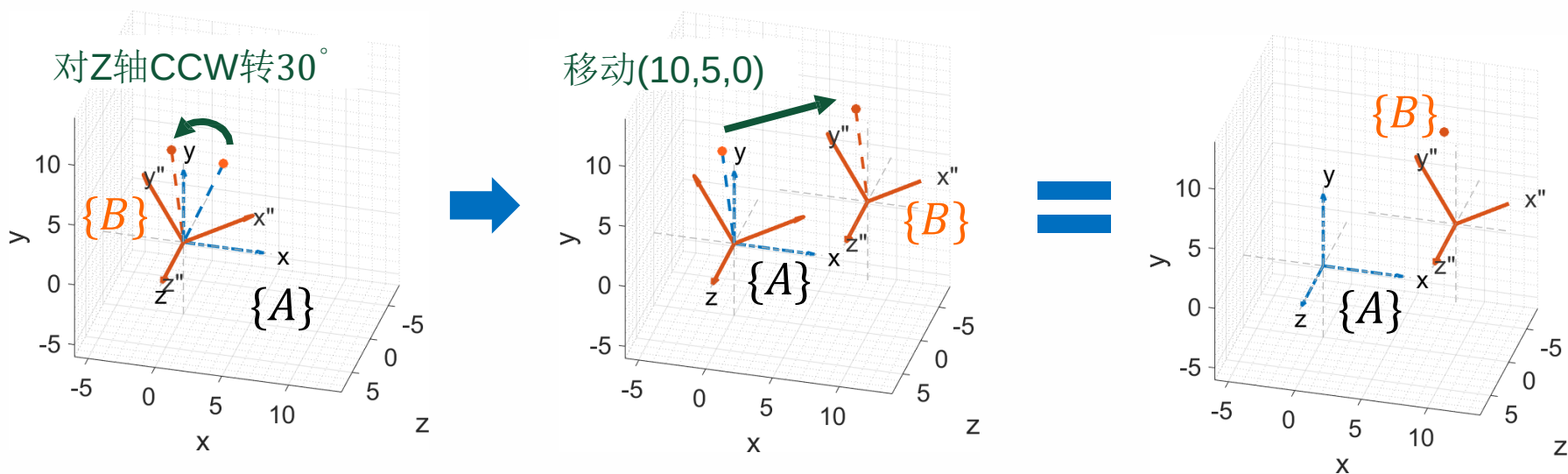
$${}^A P_2 = R_{\hat{K}}(\theta) ({}^A P_1 + {}^A Q) = R_{\hat{K}}(\theta) {}^A P_1 + R_{\hat{K}}(\theta) {}^A Q$$

Operators -3

□ Ex: Point $P_1 = \begin{bmatrix} 3 \\ 7 \\ 0 \end{bmatrix}$, 先对Z轴CCW转 30° , 然后移动 $\begin{bmatrix} 10 \\ 5 \\ 0 \end{bmatrix}$ 到 $P_2 \Rightarrow P_2 = ?$

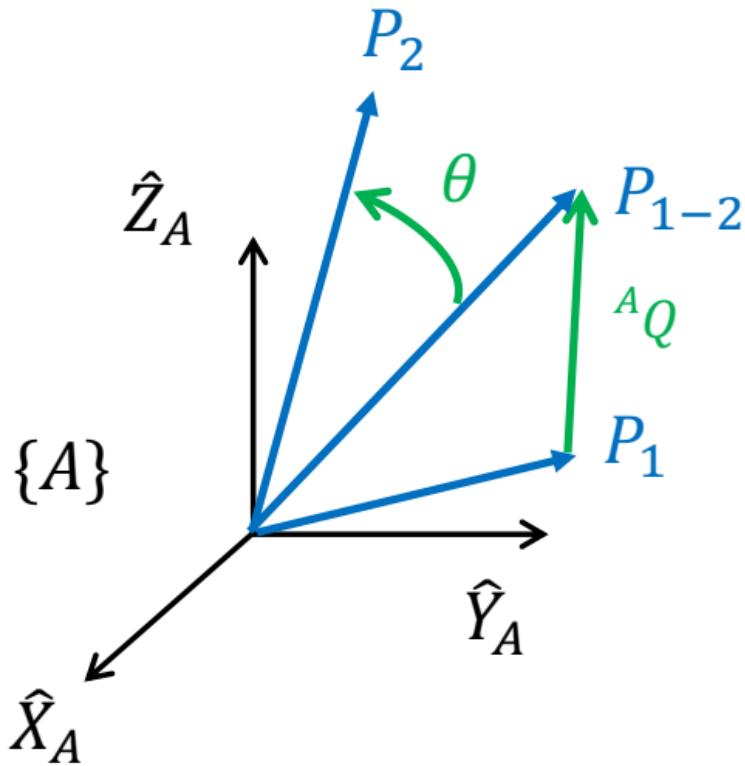
$$\begin{bmatrix} {}^A P_2 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\hat{K}}(\theta) & {}^A Q \\ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 & 10 \\ 1 & \frac{\sqrt{3}}{2} & 0 & 5 \\ \frac{1}{2} & \frac{1}{2} & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 7 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 9.098 \\ 12.562 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} | \\ {}^A P_2 \\ | \\ 1 \end{bmatrix}$$

和 “Mapping -3 “ 的答案相同, Why?



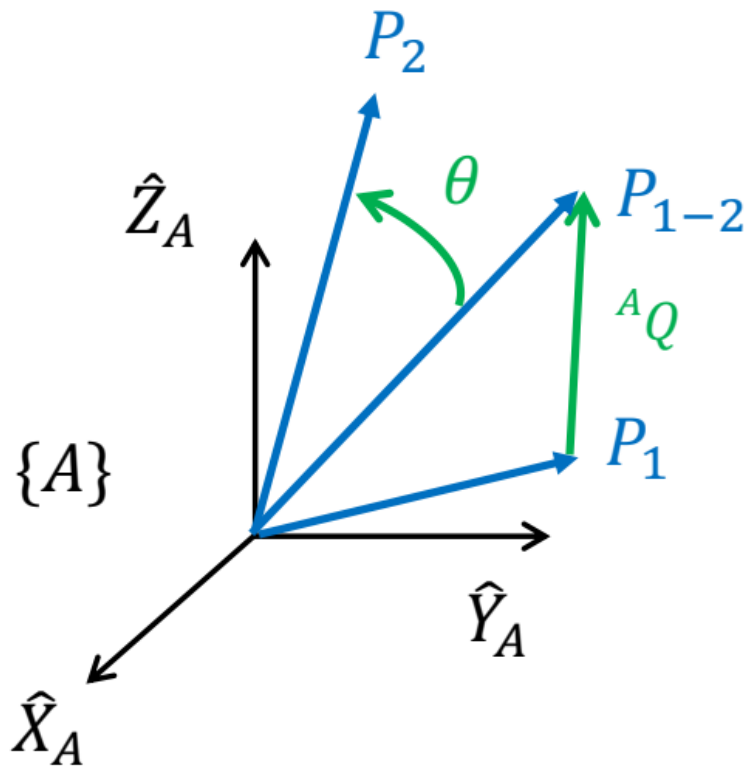
Operators -4

- In-video Quiz: 如果要如下图所示的先移动再转动，那T应该如何表达？



Operators -4

- In-video Quiz: 如果要如下图所示的先移动再转动，那T应该如何表达？

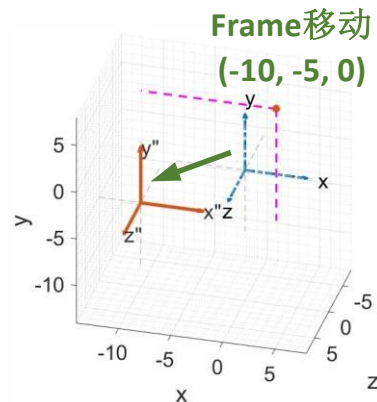
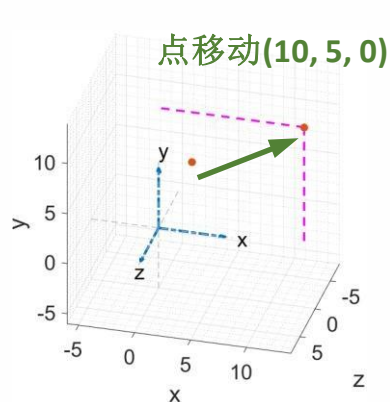


$$\begin{bmatrix} R_{\hat{K}}(\theta) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I & {}^A Q \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

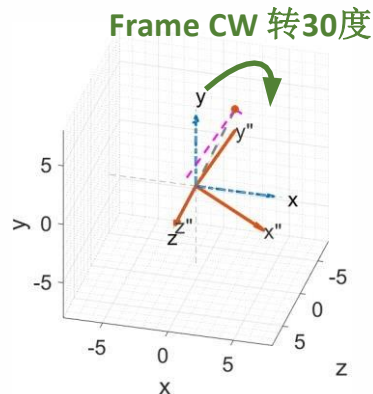
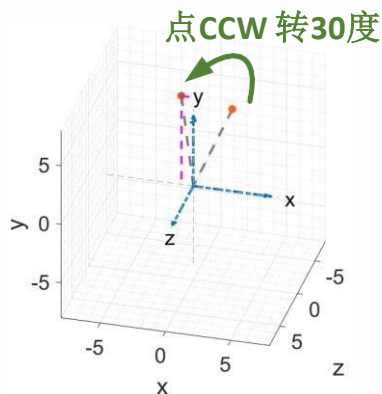
Operators -5

- 因为运动是相对的， $A_B T$ 当Operator时对向量（或点）进行移动或转动的操作，也可以想成是对frame进行“反向”的移动或转动的操作

◆ Point往前移 = frame往后移



◆ Point逆时针转 = frame顺时针转

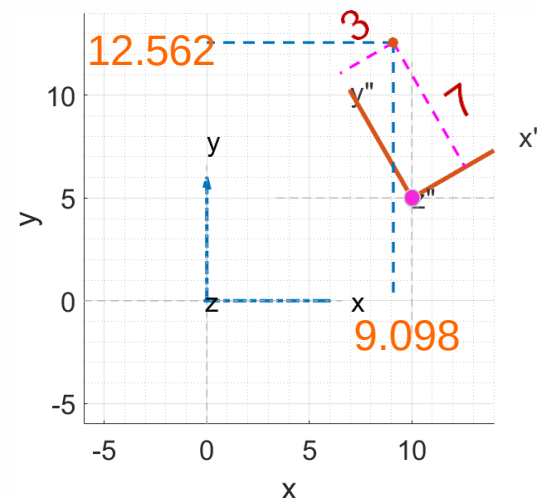
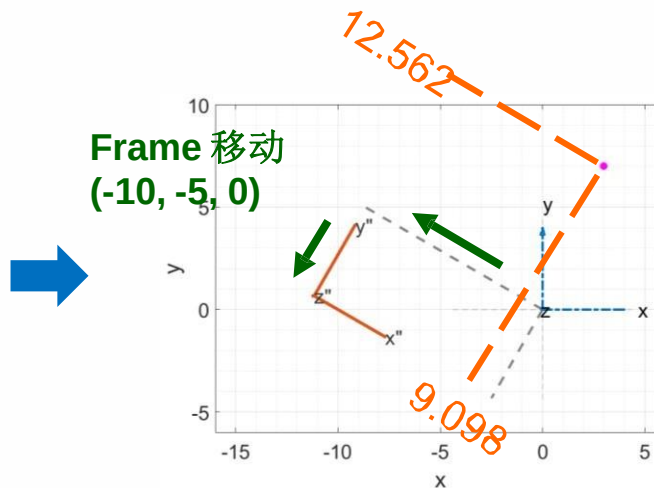
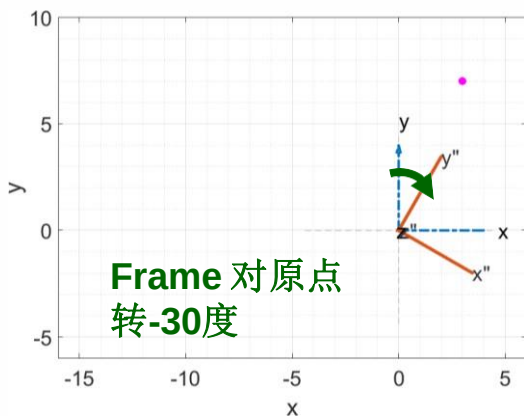


Operators -6

- Ex: Revisit Operator-3的范例，改以frame转动的角度来想

Point $P_1 = \begin{bmatrix} 3 \\ 7 \\ 0 \end{bmatrix}$, 先对Z轴CCW转 30° , 然后移动 $\begin{bmatrix} 10 \\ 5 \\ 0 \end{bmatrix}$ 到 P_2 $\Rightarrow P_2 = ?$

投影至XY平面来看



与Operator-3的
答案相同

Transformation Matrix小结

□ 齐次变换矩阵的三种用法

- ◆ 描述一个frame(相对于另一个frame)的空间状态

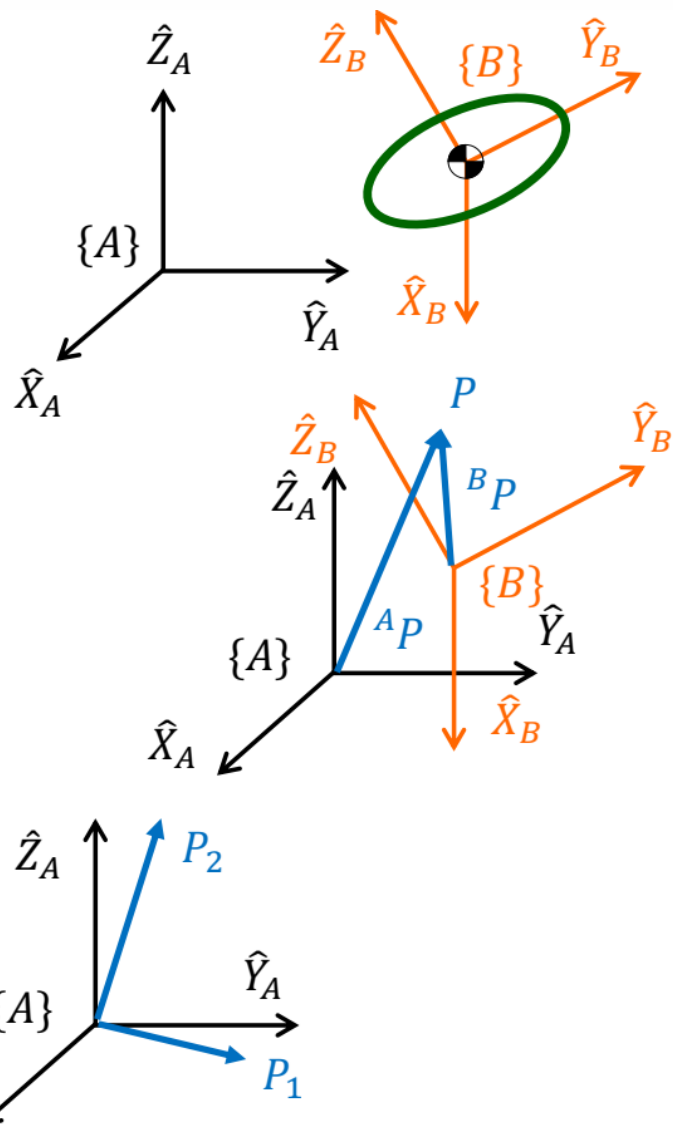
$${}^A_B T = \begin{bmatrix} | & | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B & {}^A P_{B\ org} \\ | & | & | & | \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- ◆ 将point由某一个frame的表达换到另一个frame来表达

$$\begin{bmatrix} {}^A P \\ 1 \end{bmatrix} = {}^A_B T \begin{bmatrix} {}^B P \\ 1 \end{bmatrix}$$

- ◆ 将point(vector)在同一个frame中进行移动和转动

$$\begin{bmatrix} {}^A P_2 \\ 1 \end{bmatrix} = T \begin{bmatrix} {}^A P_1 \\ 1 \end{bmatrix}$$



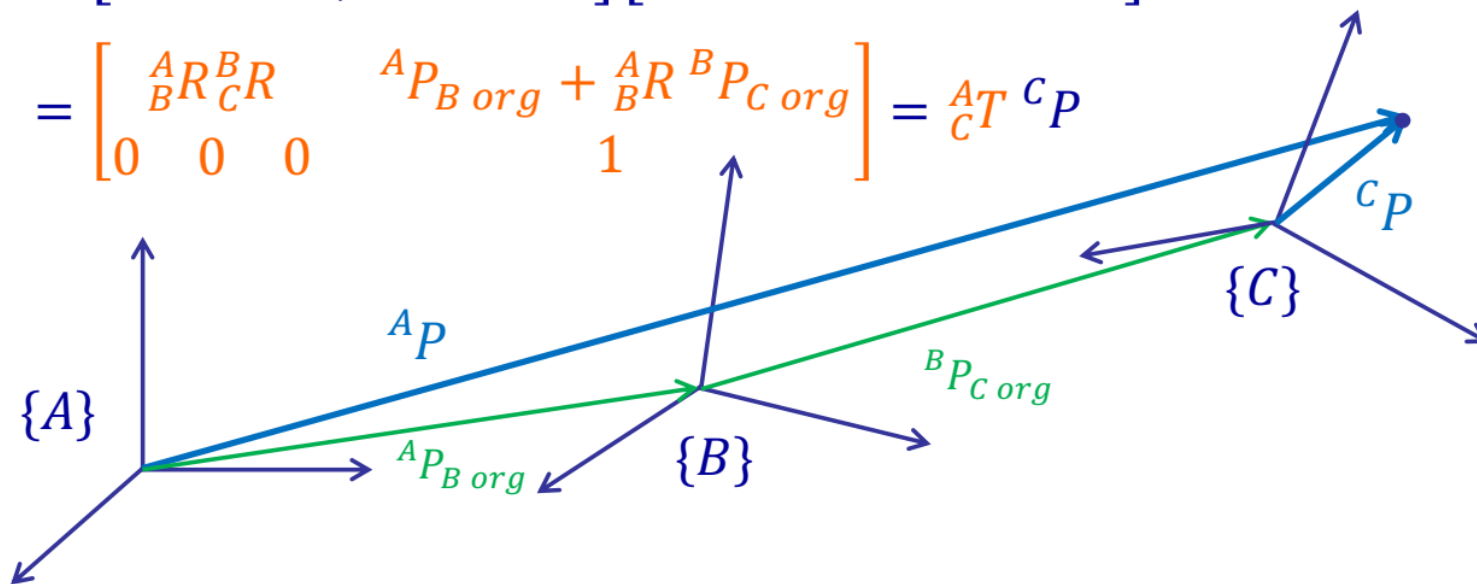
Transformation Matrix 运算-1

□ 连续运算

$${}^A P = {}^A T {}^B P = {}^A T ({}^B T {}^C P) = {}^A T {}^B T {}^C P$$

$$= \left[\begin{array}{ccc|c} {}^A R_B & & & {}^A P_{B \text{ org}} \\ \hline 0 & 0 & 0 & 1 \end{array} \right] \left[\begin{array}{ccc|c} {}^B R_C & & & {}^B P_{C \text{ org}} \\ \hline 0 & 0 & 0 & 1 \end{array} \right] {}^C P$$

$$= \left[\begin{array}{ccc|c} {}^A R_B {}^B R_C & & & {}^A P_{B \text{ org}} + {}^A R_B {}^B P_{C \text{ org}} \\ \hline 0 & 0 & 0 & 1 \end{array} \right] = {}^A T {}^C P$$



$${}^A P = {}^A T {}^B T {}^C T {}^D P$$

$$= \left[\begin{array}{ccc|c} {}^A R_B {}^B R_C {}^C R_D & & & {}^A P_{B \text{ org}} + {}^A R_B {}^B P_{C \text{ org}} + {}^A R_B {}^B R_C {}^C P_{D \text{ org}} \\ \hline 0 & 0 & 0 & 1 \end{array} \right] = {}^A T {}^D P$$

Transformation Matrix 运算-2

□ 反矩阵 ${}^A_B T = \begin{bmatrix} {}^A_B R & {}^A P_{B org} \\ 0 & 0 & 0 & 1 \end{bmatrix}$ ${}^B_A T = {}^A_B T^{-1} = ?$

$${}^A_B T {}^B_A T = {}^A_B T {}^A_B T^{-1} = I_{4 \times 4}$$

$$\begin{bmatrix} {}^A_B R & {}^A P_{B org} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^B_A R & {}^B P_{A org} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} {}^A_B R {}^B_A R & {}^A P_{B org} + {}^A_B R {}^B P_{A org} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} I_{3 \times 3} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

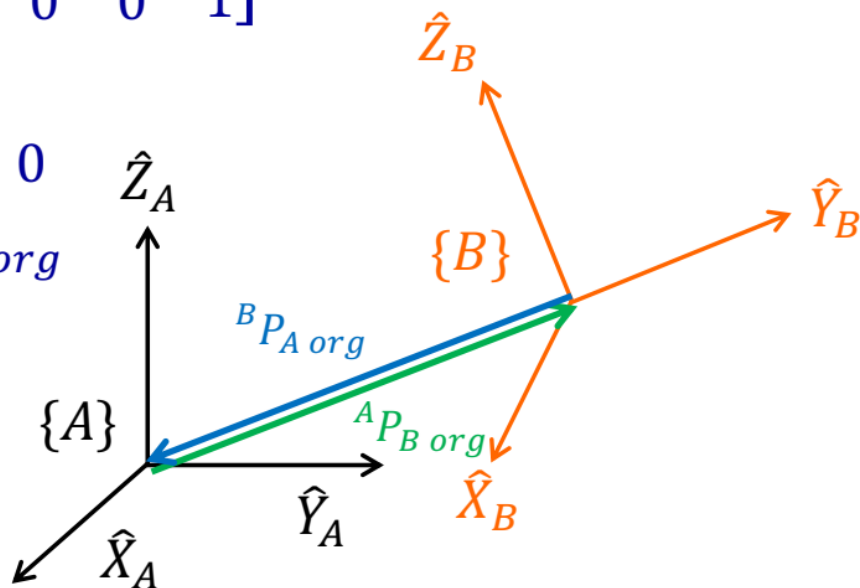
$${}^A_B R {}^B_A R = I_{3 \times 3}$$

$$\Rightarrow {}^B_A R = {}^A_B R^T$$

$${}^A P_{B org} + {}^A_B R {}^B P_{A org} = 0$$

$$\Rightarrow {}^B P_{A org} = -{}^A_B R^T {}^A P_{B org}$$

⇒ ${}^B_A T^{-1} = \begin{bmatrix} {}^A_B R^T & -{}^A_B R^T {}^A P_{B org} \\ 0 & 0 & 0 & 1 \end{bmatrix}$



Transformation Matrix 运算 -3

□ 连续运算，求未知之相对关系

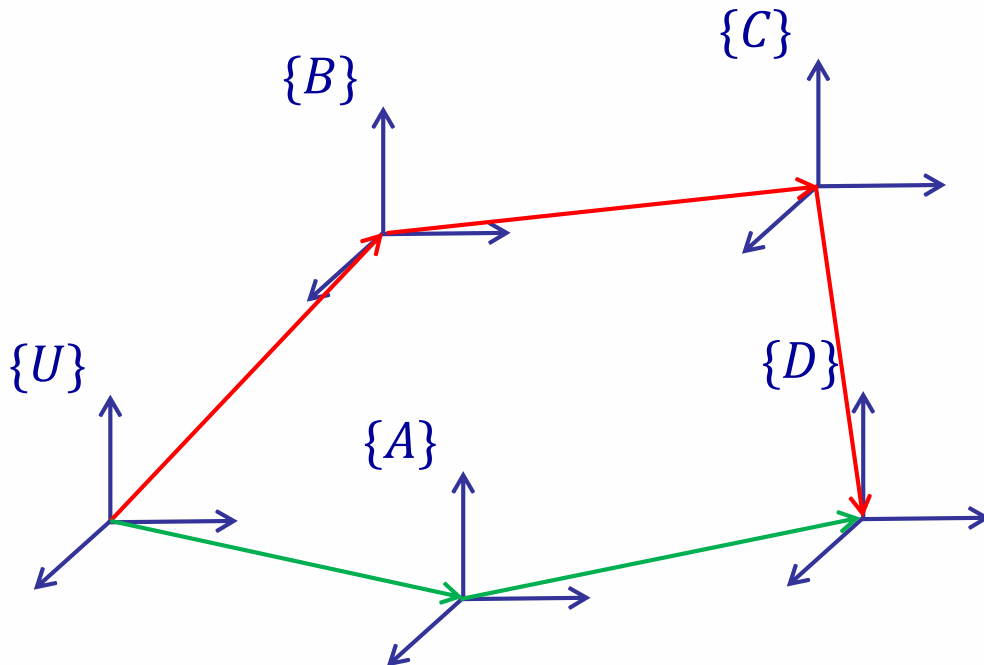
$${}^U T_D = {}^U T_A {}^A T_D = {}^U T_B {}^B T_C {}^C T_D$$

if ${}^C T_D$ unknown

$$\begin{aligned} &= ({}^U T_B {}^B T_C)^{-1} {}^U T_A {}^A T_D \\ &= {}^B T_C^{-1} {}^U T_B^{-1} {}^U T_A {}^A T_D \end{aligned}$$

if ${}^B T_C$ unknown

$$= {}^U T_B^{-1} {}^U T_A {}^A T_D {}^C T_D^{-1}$$



Transformation Matrix运算-4

□ 连续运算 法则

◆ Initial condition: $\{A\}$ and $\{B\}$ coincide ${}^A_B T = I_{4 \times 4}$

◆ $\{B\}$ 对 $\{A\}$ 的转轴旋转：用“premultiply”

◦ 以operator来想，对某一个向量，“以同一个坐标为基准”，进行转动或移动的操作

◦ Ex: $\{B\}$ 依序经过 T_1 、 T_2 、 T_3 三次transformations

$${}^A_B T = T_3 T_2 T_1 I \quad v' = {}^A_B T v = T_3 T_2 T_1 v$$

◆ $\{B\}$ 对 $\{B\}$ 自身的转轴旋转：用“postmultiply”

◦ 以mapping来想，对某一个向量，从最后一个frame“逐渐转动或移动”来回到第一个frame

◦ Ex: $\{B\}$ 依序经过 T_1 、 T_2 、 T_3 三次transformation

$${}^A_B T = I T_1 T_2 T_3 \quad {}^A_B P = {}^A_B T {}^B_B P = I T_1 T_2 T_3 {}^B_B P$$

Transformation Matrix运算-5

□ 连续运算 小结

- ◆ 以固定的{A}或移动的{B}为基准进行移动换转动操作，
transformation matrix应用不同的连乘方式
- ◆ 思考逻辑和考量固定角 vs. 欧拉角的连续旋转顺序相似